



Global Cities' Cycling Initiatives

Emerging & Progressive Practices from the Global South

NACTO-Global Designing Cities Initiative

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Velo-City, Dublin, Ireland

27 June 2019

@GlobalStreet
www.globaldesigningcities.org

NACTO National Association of City
Transportation Officials
GDCI Global Design Cities Initiative

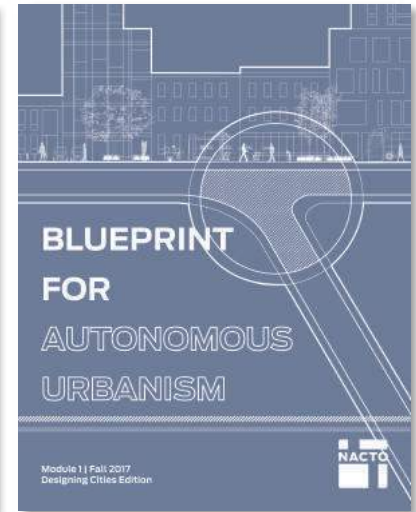
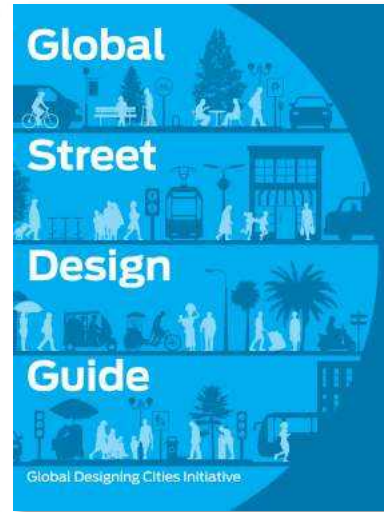
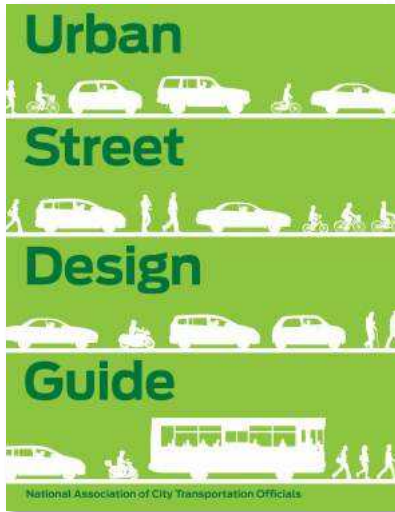
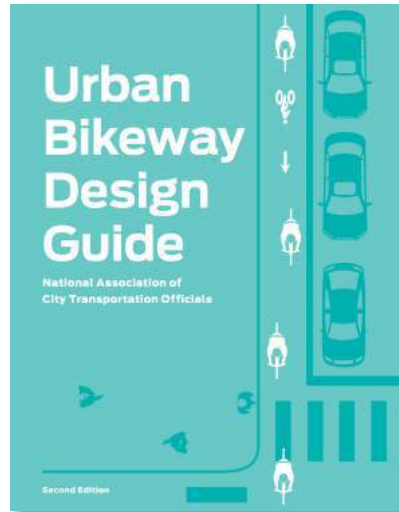


**Global
Designing
Cities
Initiative**



**Janette
Sadik-Khan**

NACTO Design Guides





**Global
Designing
Cities
Initiative**

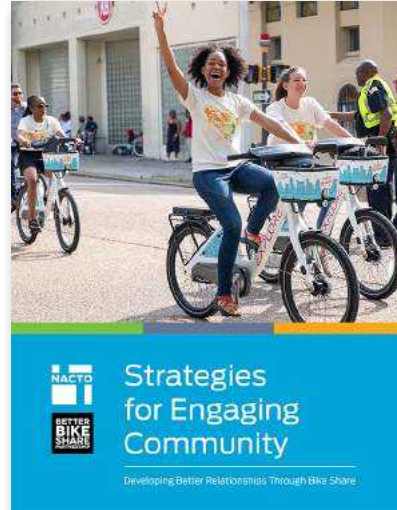


**Bloomberg
Philanthropies**
INITIATIVE FOR GLOBAL ROAD SAFETY

Guide

Global Designing Cities Initiative

NACTO Supplements



Designing Streets for Kids



Global Designing Cities Initiative

GSDG Supplement

A New Approach to Street Design



Design for ALL street users



Pedestrians



Cyclists



Transit Riders



Motorists

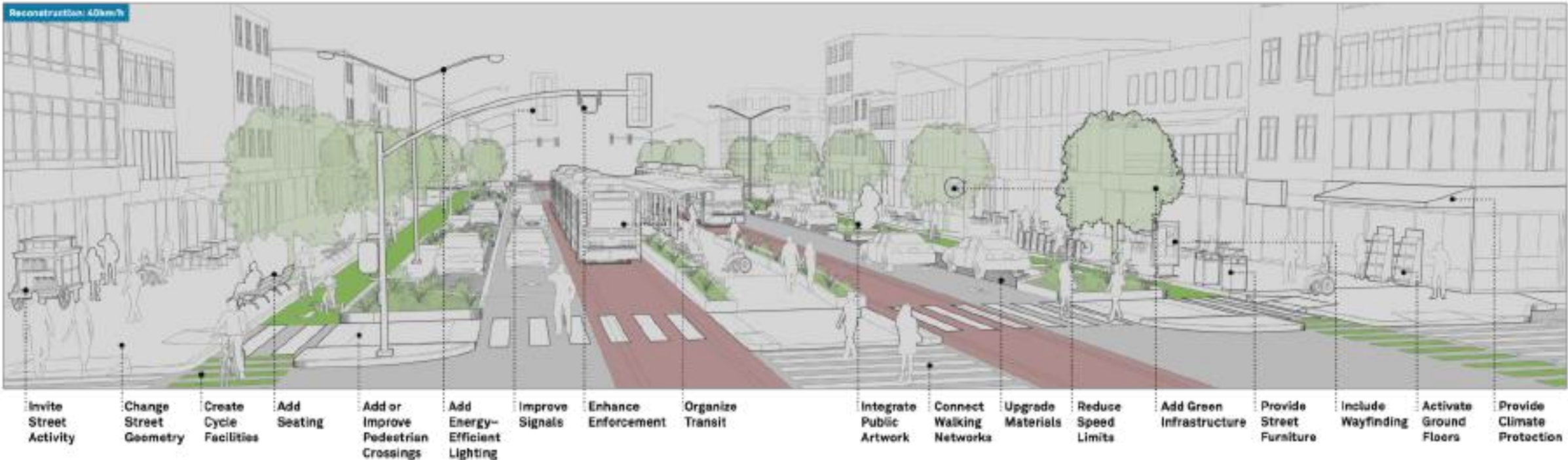


**Freight Operators
and Service
Providers**



**People Doing
Business**

What is possible?



Geometries and dimensions

**Pedestrians**

**Cyclists**

**Transit Riders**

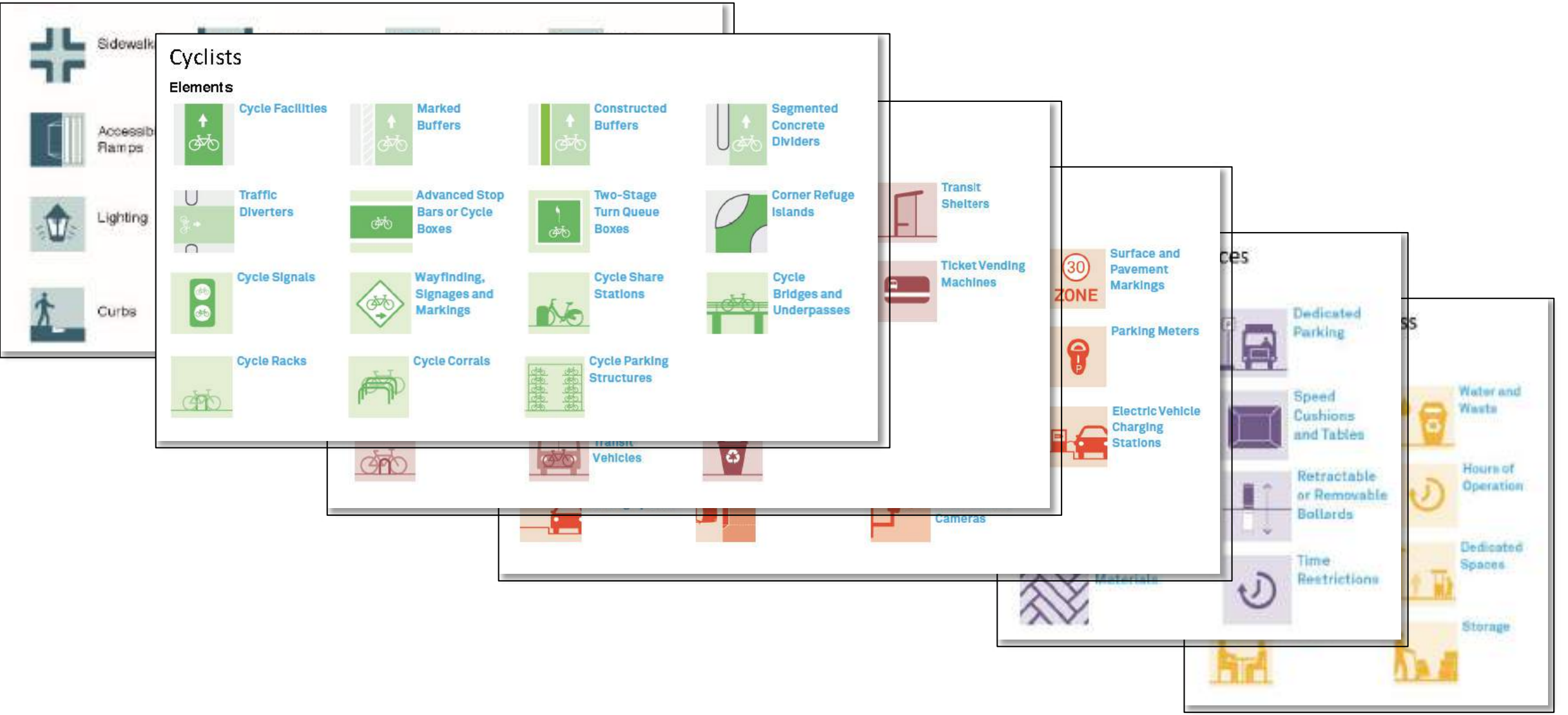
**Motorists**

**Freight Operators and Service Providers**

**People Doing Business**



Elements and features



How to approach at the network scale



Pedestrians



Cyclists



Transit Riders



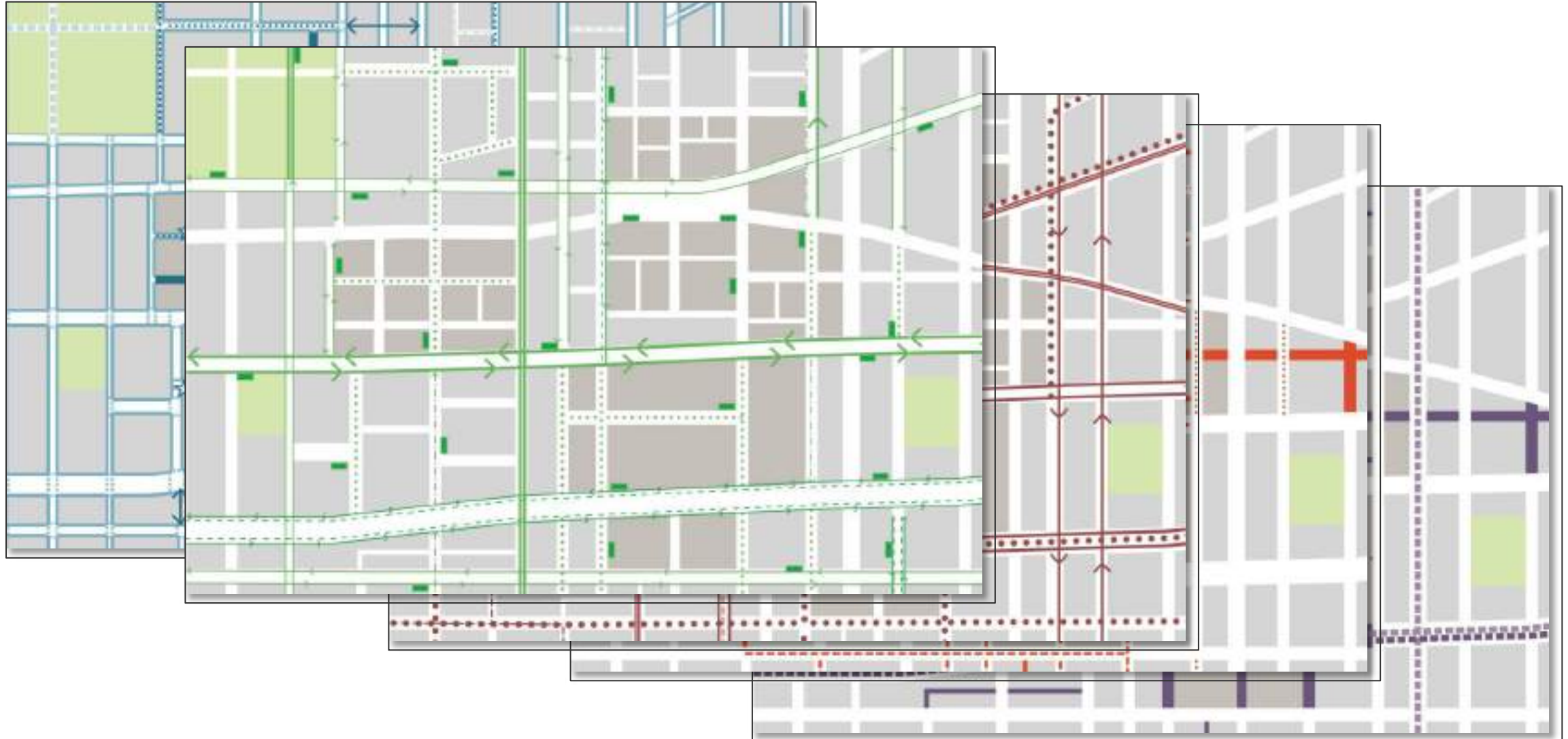
Motorists



Freight Operators
and Service
Providers



People Doing
Business



Cyclists

Redefine the Cyclist



Cyclists

Designing for All Ages and Ability



Cyclists

Types: Speed, Variations and Dimensions

Children and families

Commuter

Recreational cyclists



0 km/h

10 km/h

20 km/h

> 30 km/h

Interested but concerned

Confident

Fearless

Cycle Facilities



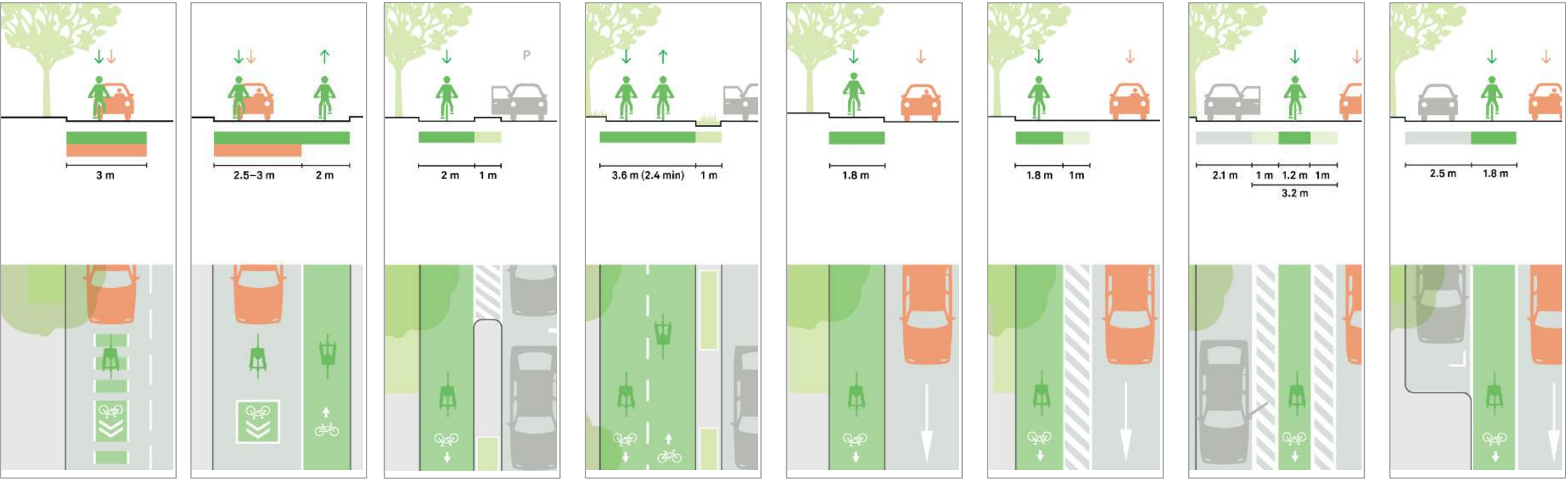
Curb Zone

Cycle Lane

Buffer Zone

Cycle Facilities

Geometry



Cycle Street

Contraflow
Cycle Street

Protected
Cycle Track

Bidirectional
Cycle Track

Raised
Cycle Track

Curbside
Buffered
Cycle Lane

Buffered
Cycle Lane

Conventional
Cycle Lane

Bicycle Networks

Key Network Considerations



- Safe
- Connected
- Comfortable

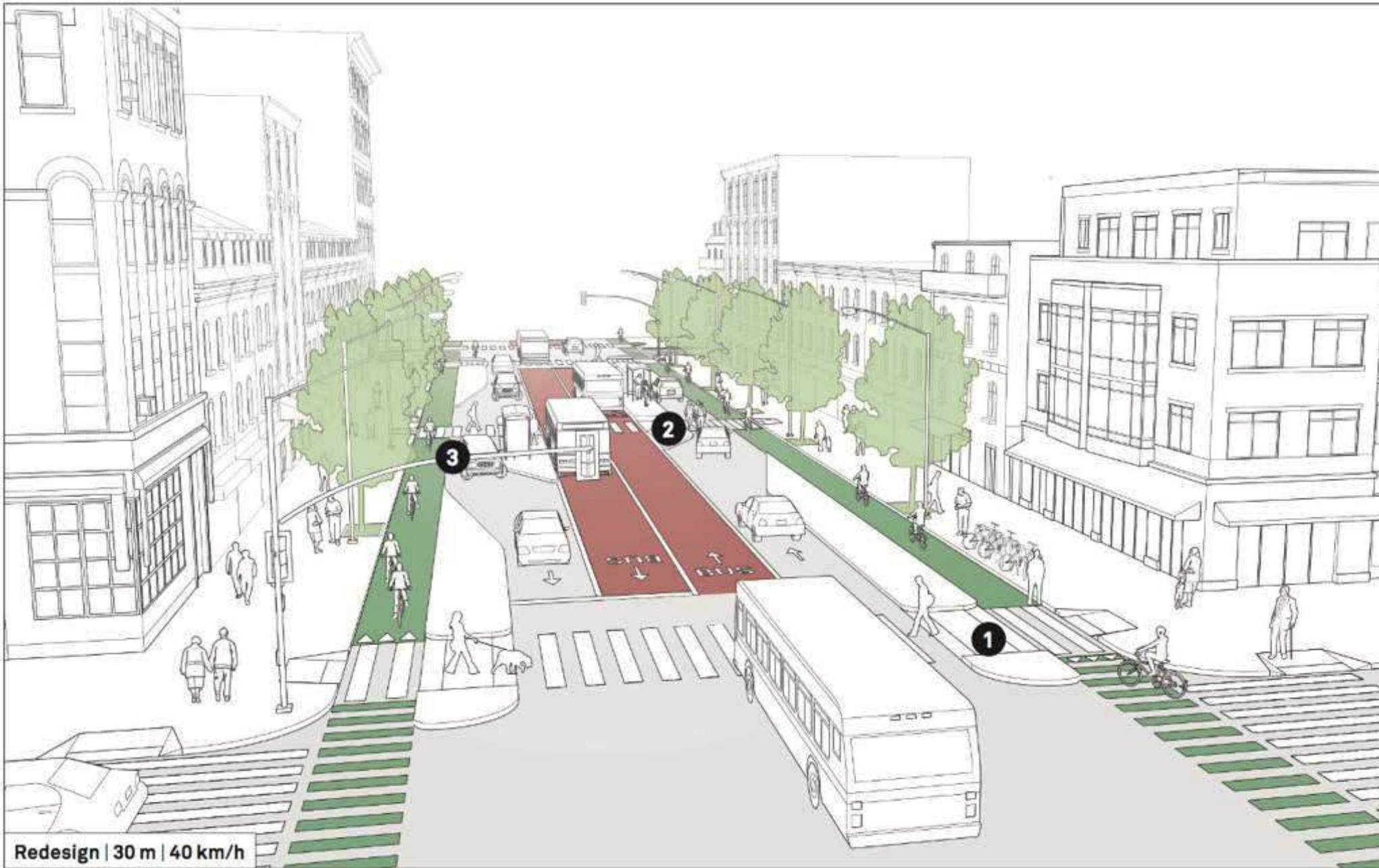
- Cycle street
- > Cycle lane
- .-.-> Contraflow cycle lane
- - -> Buffered cycle track
- > One-way cycle lane
- ===== Two-way cycle track
- Cycle share station

Central Two-Way Streets



Before

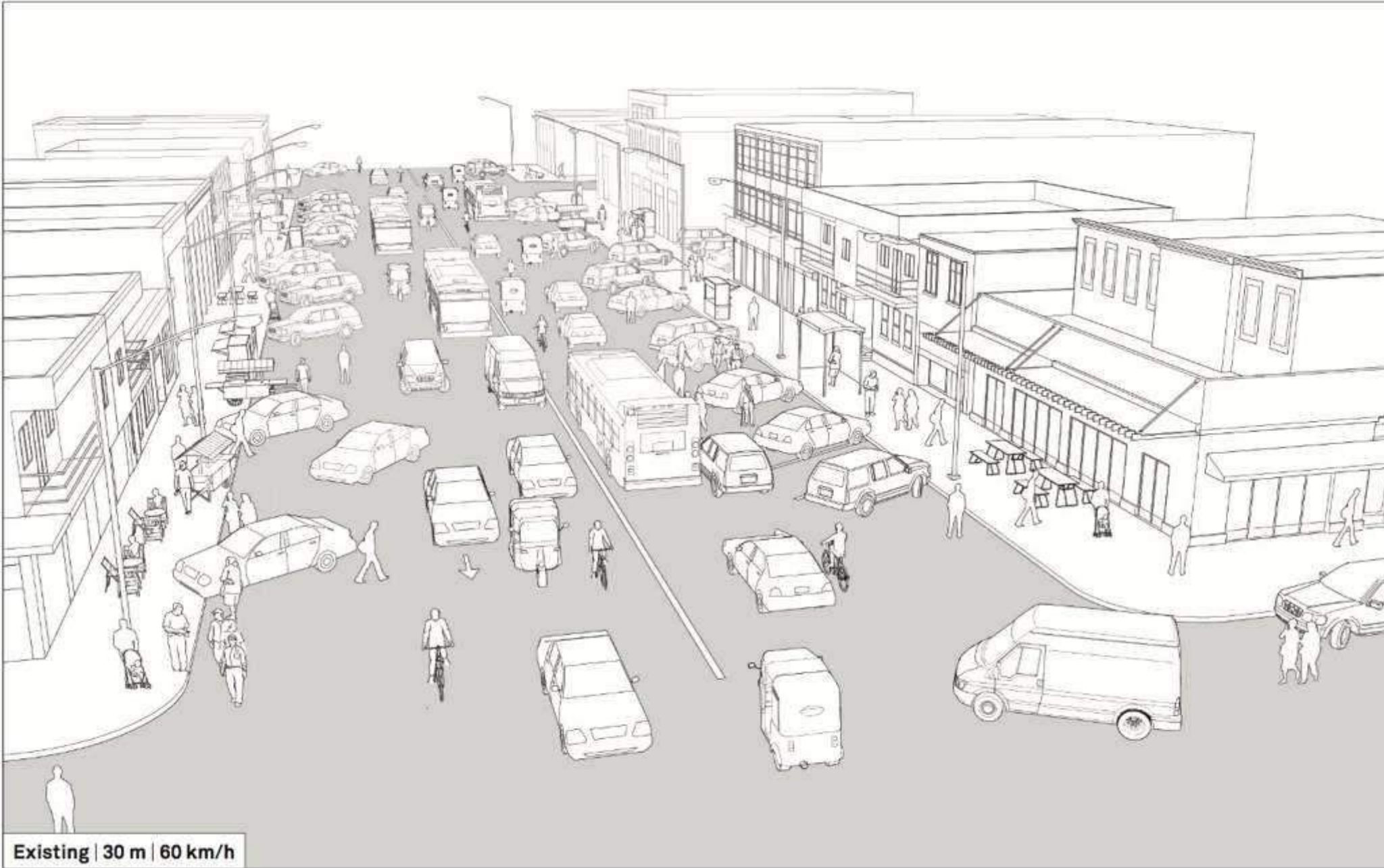
Central Two-Way Streets



Redesign | 30 m | 40 km/h

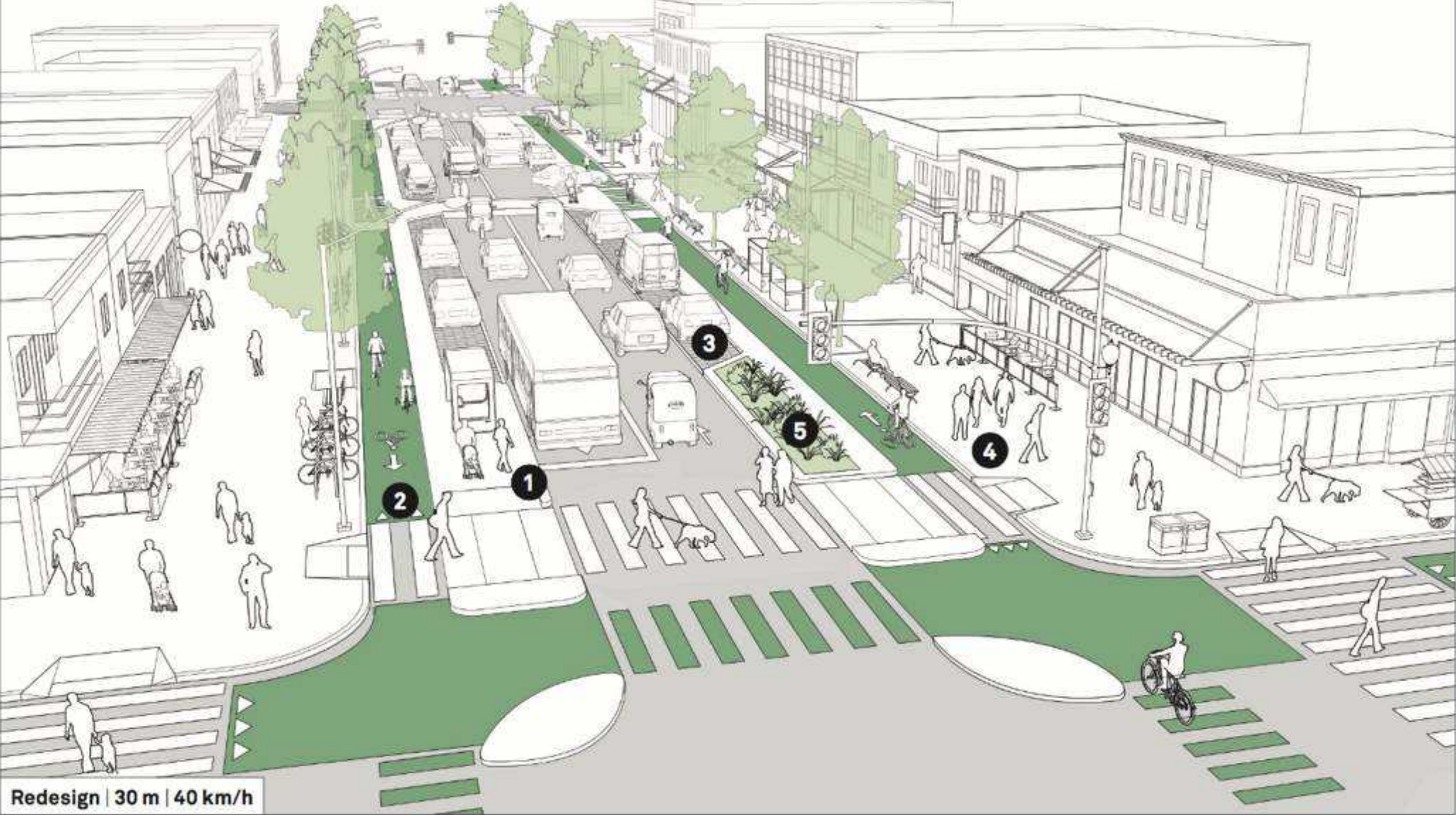
After

Neighborhood Main Streets

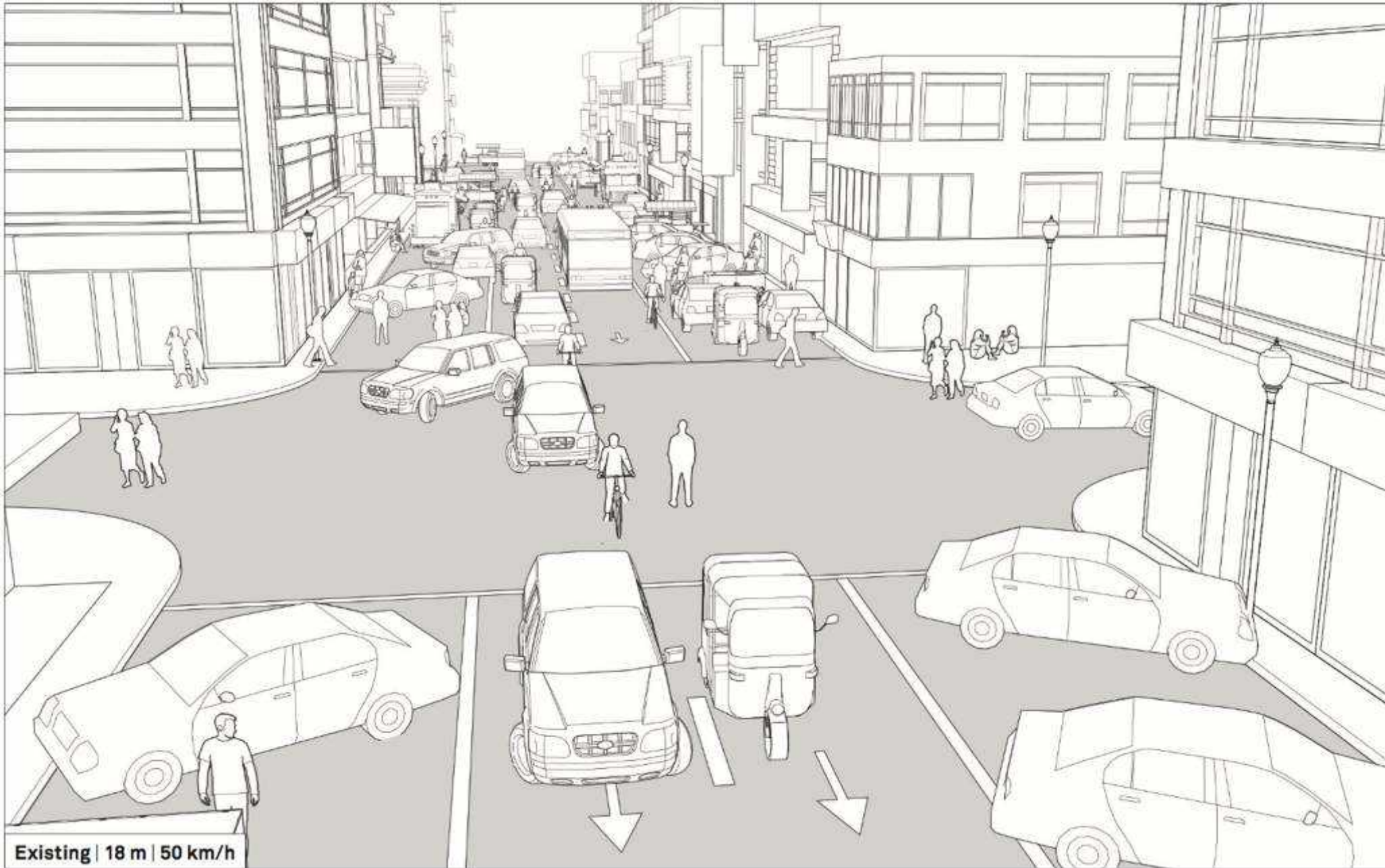


Before

Neighborhood Main Streets



Residential Streets

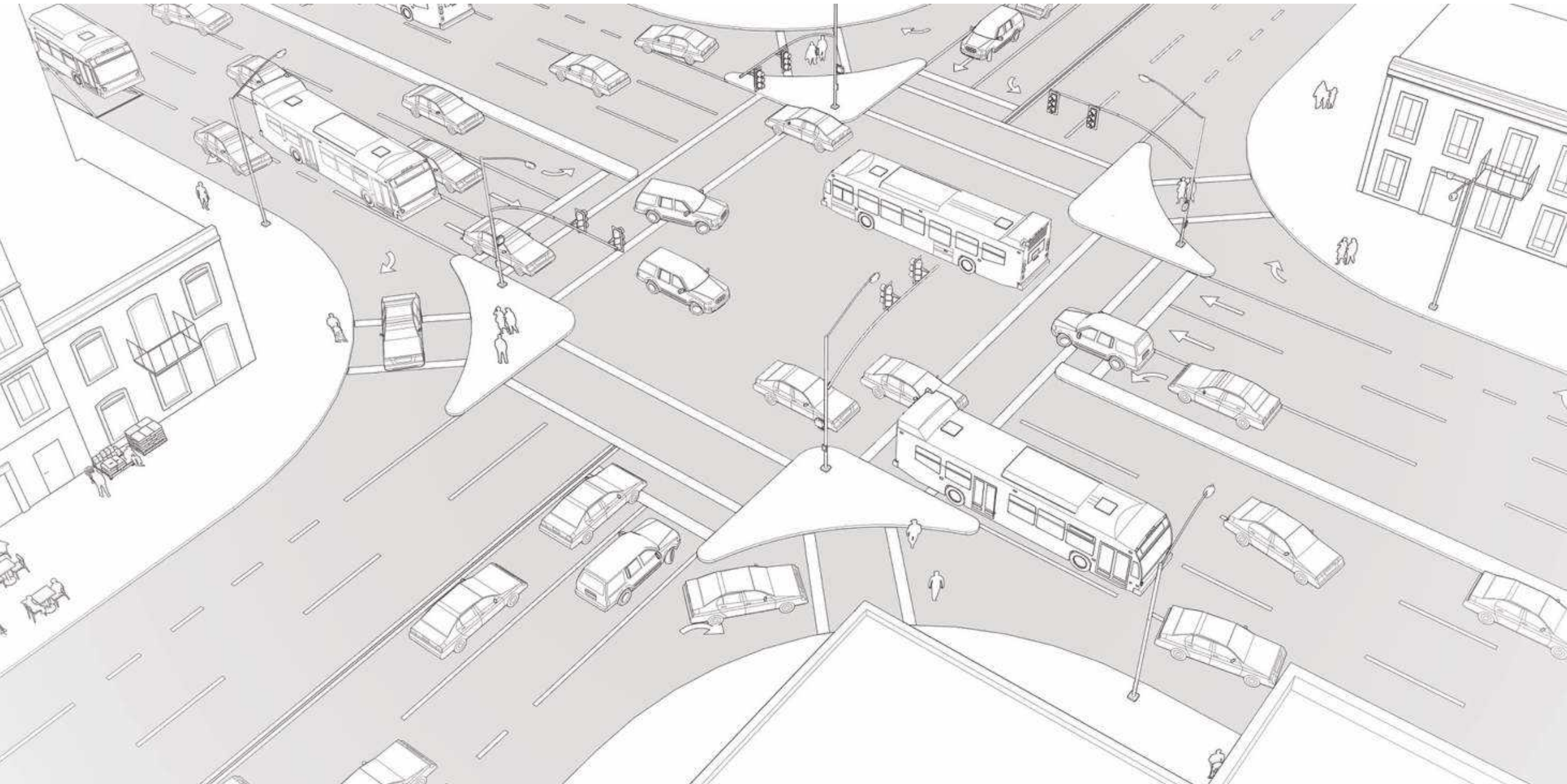


Local Streets

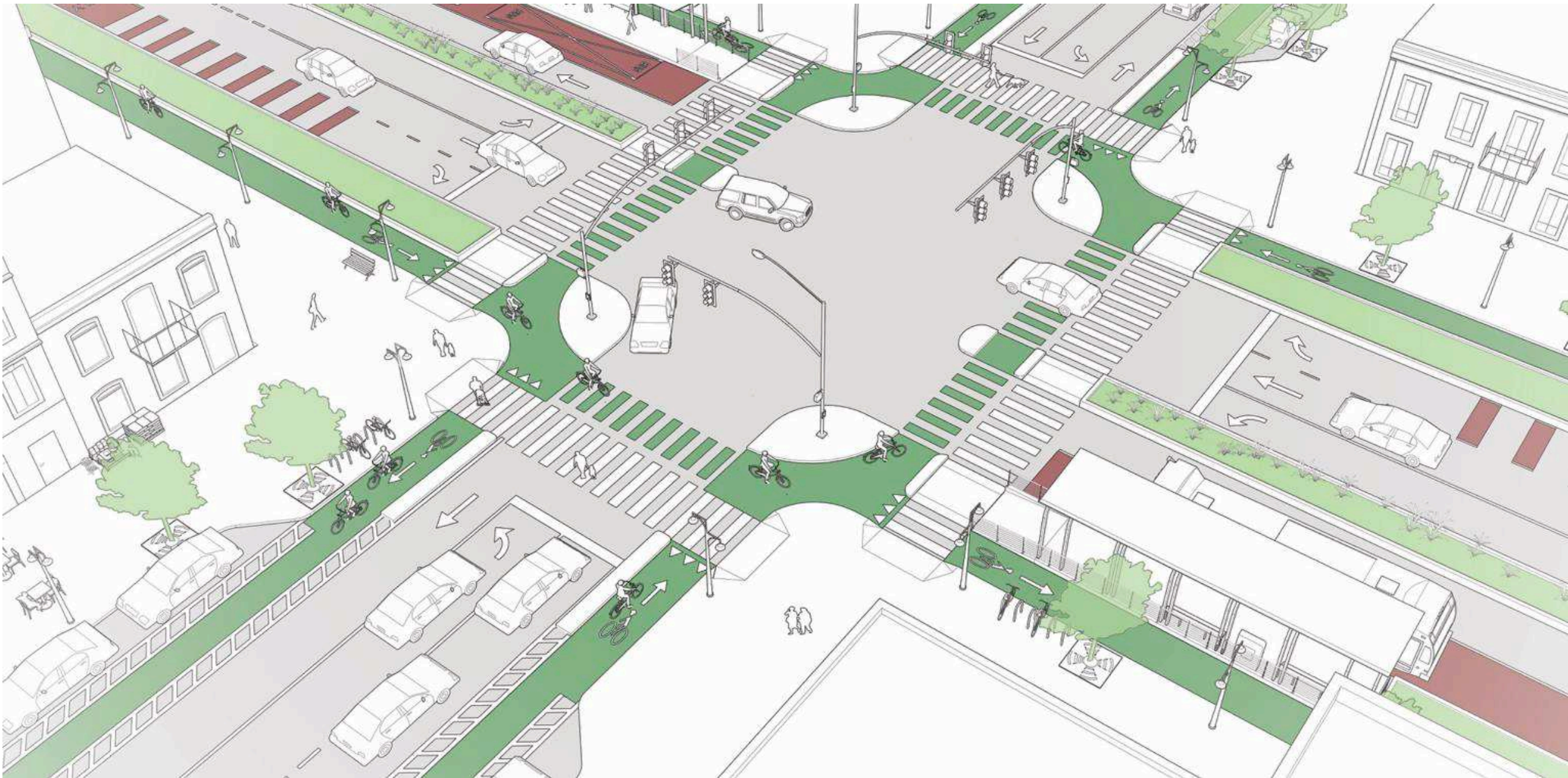


After

Protected Intersections



Protected Intersections



Addis Ababa Ethiopia



Fortaleza Brazil



Bogotá Colombia



São Paulo Brazil



Mumbai India



Global
Designing
Cities
Initiative



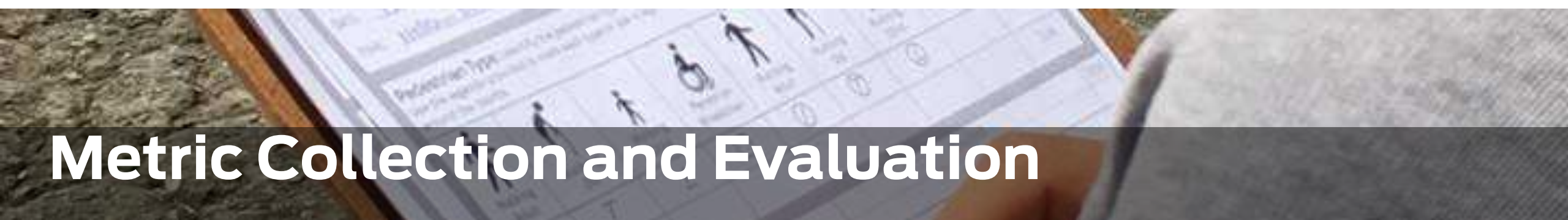
Policy and Design Guidance



Capacity Building and Community Engagement



Interventions and Transformations



Metric Collection and Evaluation

Addis Ababa Ethiopia



Fortaleza Brazil



Bogotá Colombia



Addis Ababa Ethiopia



Fortaleza Brazil



Bogotá Colombia



**Emerging
Cycling
Movement**

Addis Ababa Ethiopia



Fortaleza Brazil



Bogotá Colombia



**Emerging
Cycling
Movement**

**Progressing
Cycling
Momentum**

Addis Ababa Ethiopia



Fortaleza Brazil



Bogotá Colombia



**Emerging
Cycling
Movement**

**Progressing
Cycling
Momentum**

**Reinforcing
Cycling
Culture**

Addis Ababa

(Ethiopia)

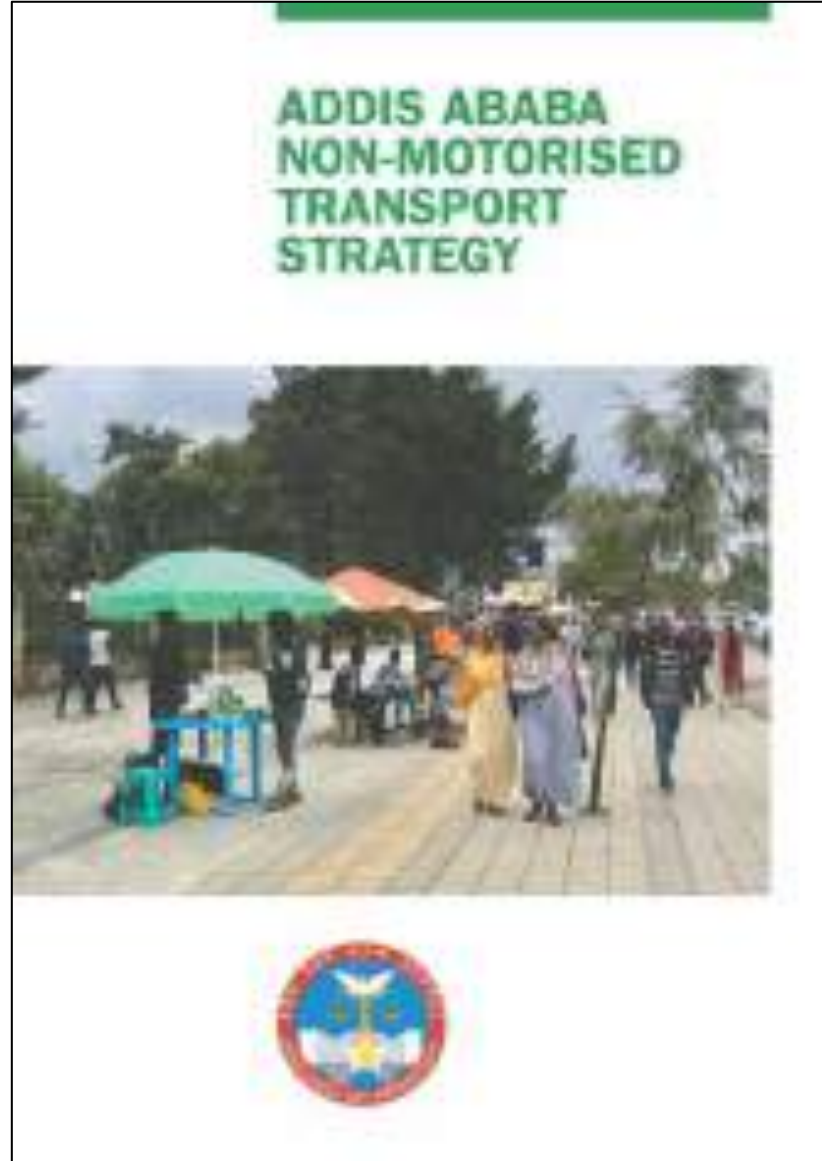


Objective:
60% of total
trips through
NMT by **2021**

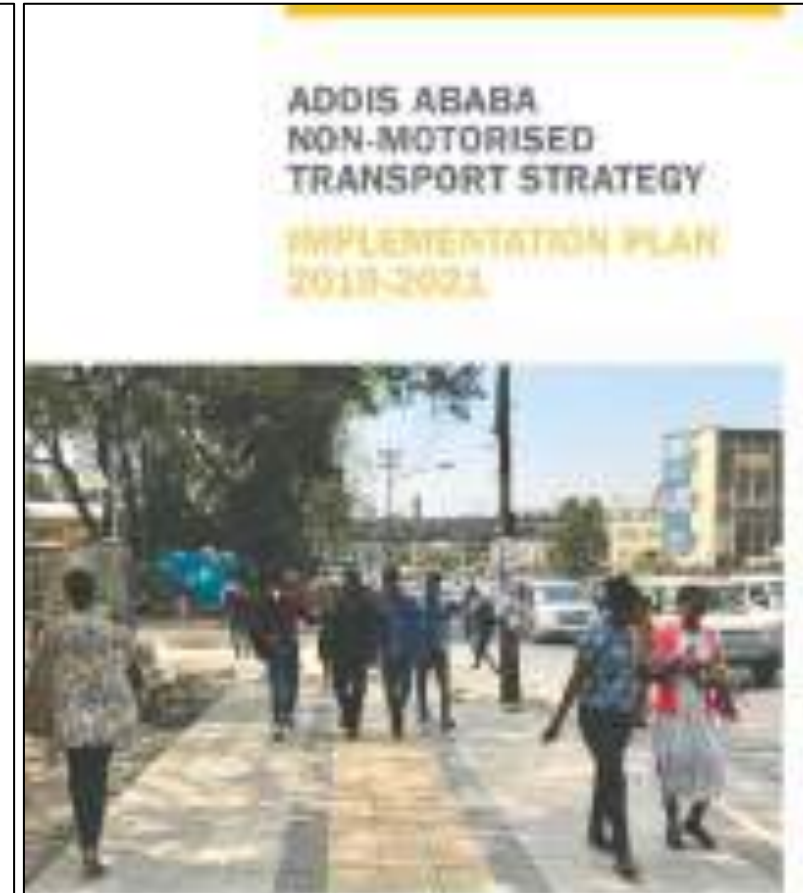
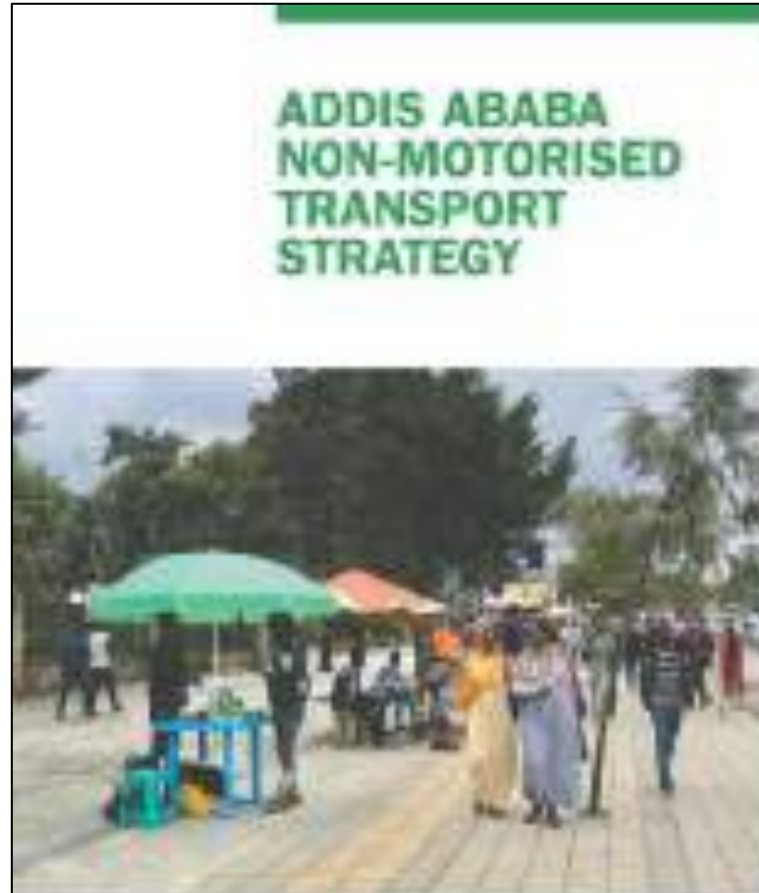
The background image shows a person wearing a yellow jacket and a brown hood, painting the word 'CYCLING' in white chalk on a dark asphalt road. The person is seen from the side, and their hand is visible as they paint. The word 'CYCLING' is written in large, bold, capital letters. The image is partially obscured by a blue text box on the right side.

Objective:
200 kms of
cycling facilities
by **2028**

Visions and Goals



Action Items and Strategies



Addis Ababa | Ethiopia

Open Streets Days as a Catalyst for Cycling



መንገድ ለሰው!

ከተሽከርካሪ ነጻ ቀን

ግንቦት 4 ቀን 2011 ዓ.ም ለ6ኛ ጊዜ ይከበራል

ከብሔራዊ ቲያትር - ሜክሲኮ አደባባይ - ልደታ ፍርድ ቤት ከተሽከርካሪ ነጻ ይሆናሉ ሰዓት: ከጠዋቱ 1:30 እስከ ከቀኑ 6:00

በዚህ ዝግጅት

- የአካል ብቃት እንቅስቃሴዎች ይደረጋሉ፤
- የእግር ጉዞ ይከናወናል፤
- የብስክሌት ውድድር ይደረጋል፤
- የሰርከስና የባህል ትርዒቶች ይቀርባሉ፤
- የእግር ኳስ ውድድር ይከናወናል፤
- የስኬቲንግ ውድድር ይደረጋል፤
- በፍላጎት ላይ የተመሰረተ የጤና ምርመራዎች ይደረጋሉ፤

በዚህ ዝግጅት ላይ ተሳታፊ እንዲሆኑ ተግባራዊ!



የአዲስ አበባ ከተማ አስተዳደር መንገድ እና ትራንስፖርት ቢሮ
ADDIS ABABA CITY ADMINISTRATION ROAD AND TRANSPORT BUREAU



ትፖሙ
የፈጠረው ጥራት ማስተላለፊያ ይሰጣል
Transport Programs Management Office



ITDP
Institute for Transportation & Development Policy
Promoting sustainable and equitable transportation worldwide.



የአዲስ አበባ ከተማ መንገድ ትራፊክ ማኔጅመንት ኤጀንሲ
Addis Ababa City Road Traffic Management Agency

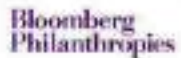


መንገድ ከሰው!

ከመኪና ነጻ
በሆነ
መንገድ
እንንቀሳቀስ!

ግንቦት 4/2011
12 ሰዓት - 7 ሰዓት

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ከሜክሲኮ - ብሔራዊ
ከቤቴል - ኪዳነ ምኅረት
ከሳፋሪ - ፊጋ
ከለቡ መብራት - ጀም 1
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Bloomberg Philanthropies



የአዲስ አበባ ከተማ አስተዳደር



የአዲስ አበባ ከተማ መንገድ ትራፊክ ማኔጅመንት ኤጀንሲ



ITDP



WORLD RESOURCES INSTITUTE

Addis Ababa | Ethiopia

Open Streets Day – Bike Training for Kids



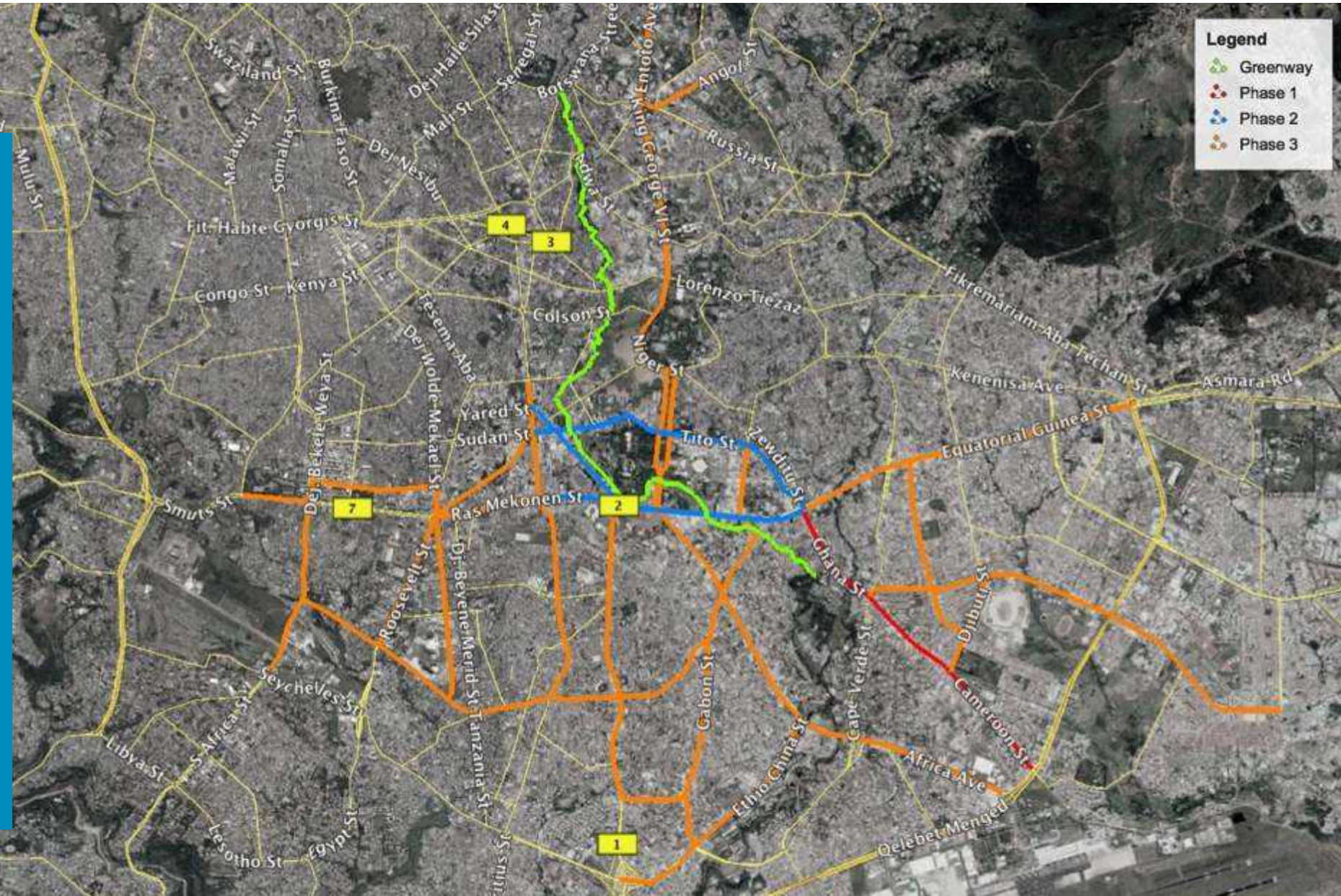
Addis Ababa | Ethiopia

Open Streets Day – Cycle Safety for All Ages



Development of Policies and Framework

2020: 10km
2021: 20km
2022: 30km
.
.
.
2028: 200



Addis Ababa | Ethiopia

Cycling Infrastructure vs. Complete Street Transformations

Difficult to fit in a successful bike lane on many of Addis Ababa's streets, since all-round improvements to sidewalks, road surface, and parking are needed.

Implementation of a cycle network requires complete street transformations on many roads or sections thereof.



Joseph Tito St.

Addis Ababa | Ethiopia

Identified Street Typologies

1. THROUGH ROAD

- Separation of traffic modes
- Reduce conflicts
- Safe provisions at intersections



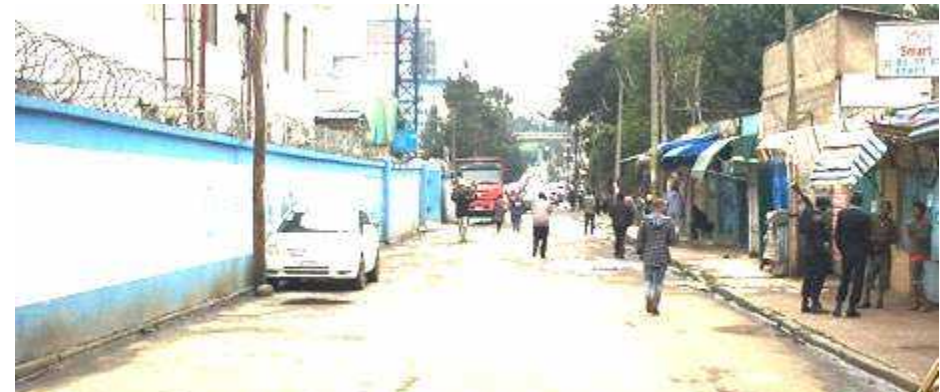
2. CONNECTOR ROAD

- Some separation of traffic modes
- Traffic calming at intersections
- Lower speed limits



3. LOCAL STREET

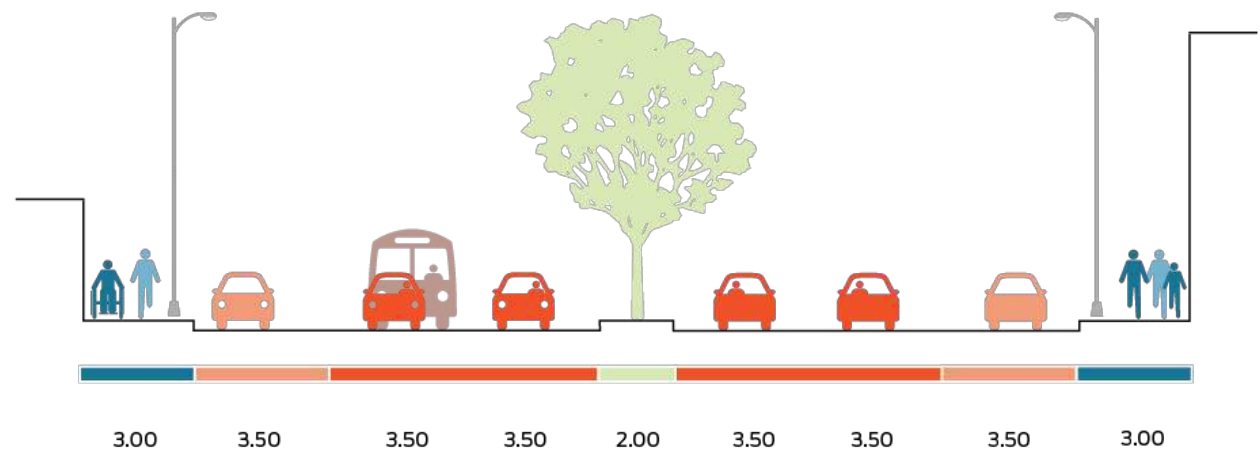
- Mixing of traffic modes
- Traffic calming
- On-street parking
- Very low speed limits



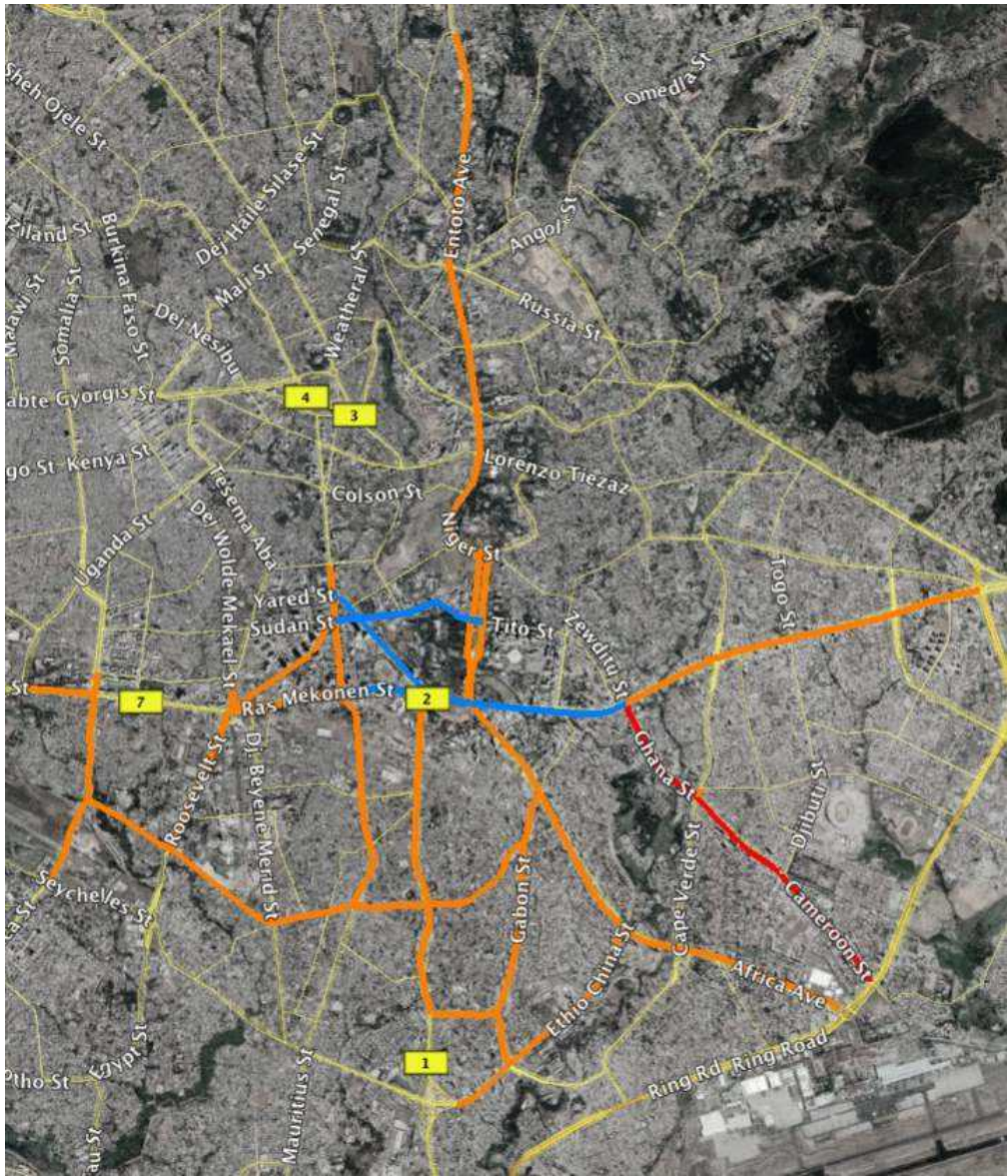
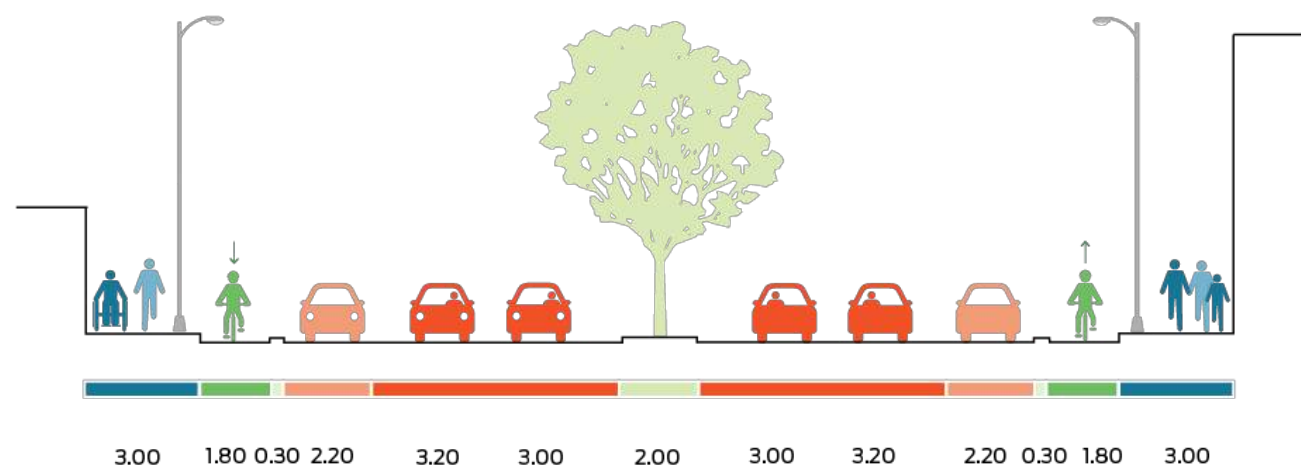
Addis Ababa | Ethiopia

Cycle corridors on Through Roads (~30-40m)

EXISTING:



PROPOSED:





Through Road | Vision | Before

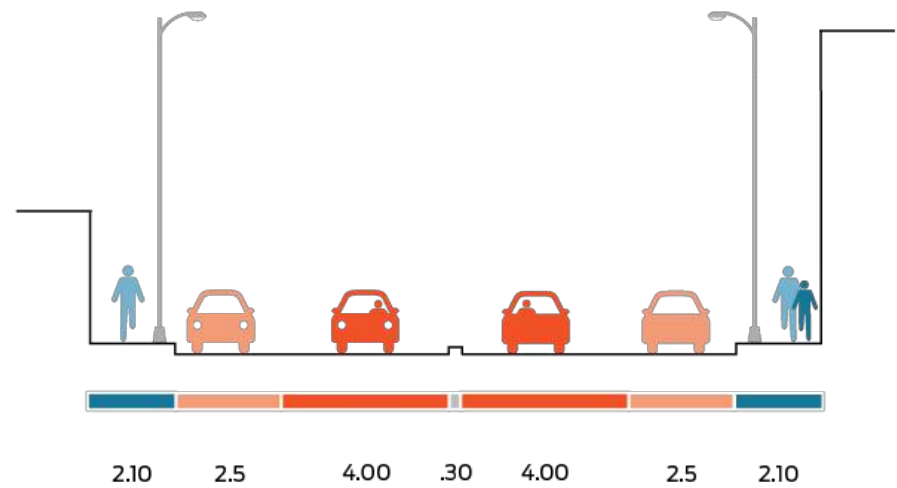


Through Road | Vision | After

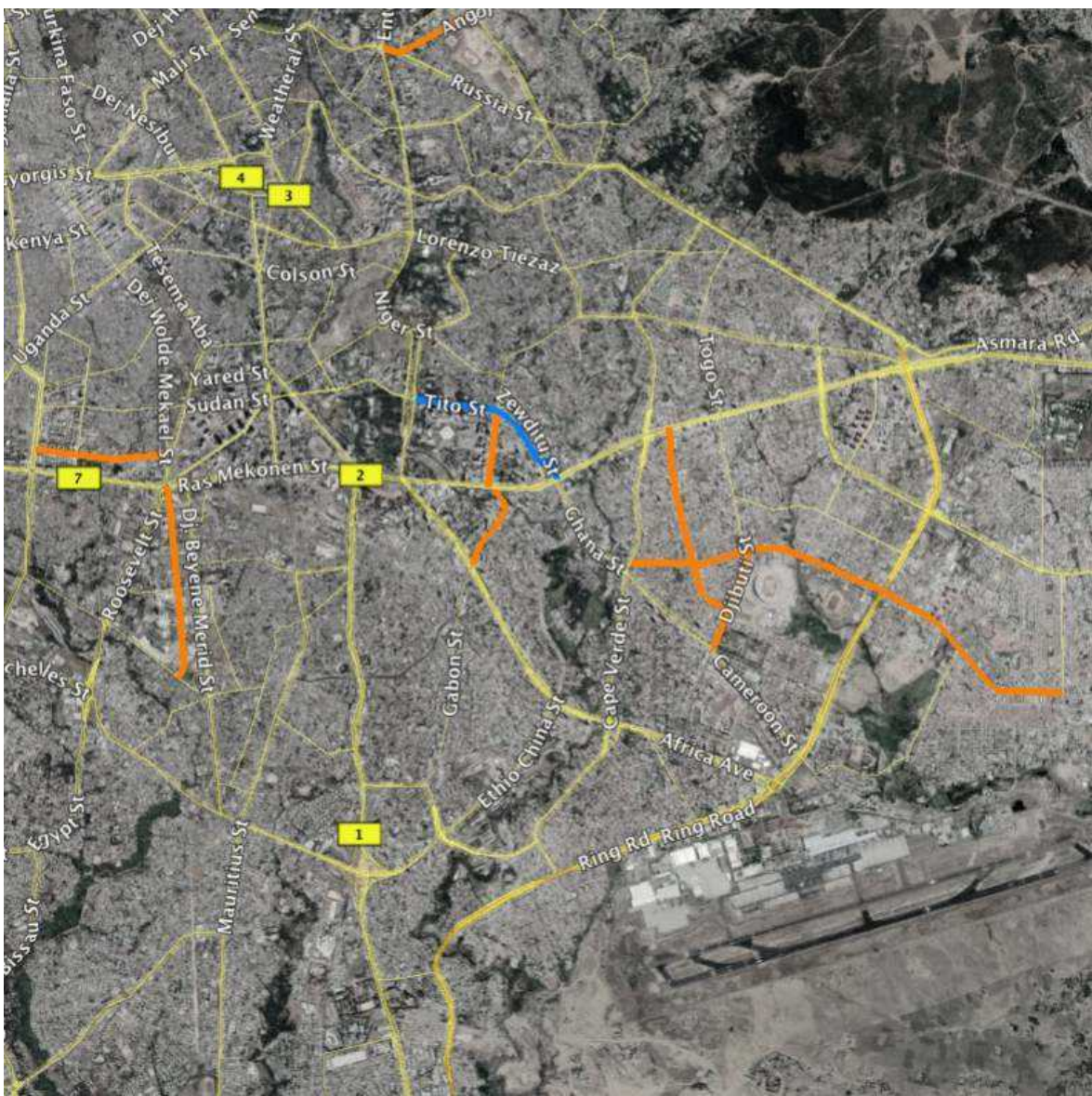
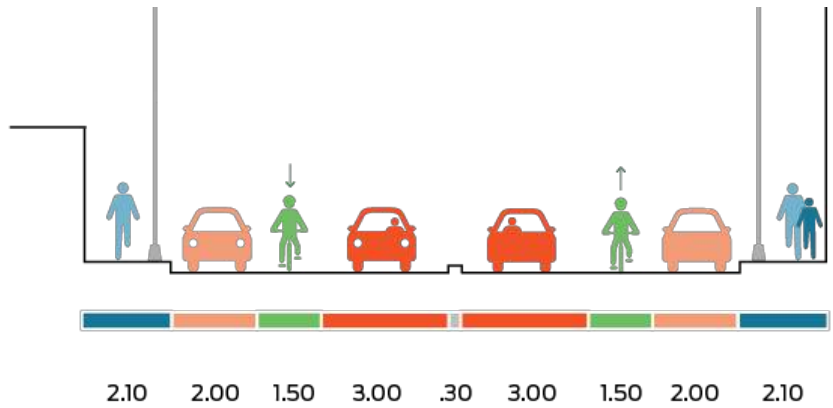
Addis Ababa | Ethiopia

Cycle corridors on Connector Streets(~18-30m)

EXISTING:



PROPOSED:





Connector Street| Vision | Before

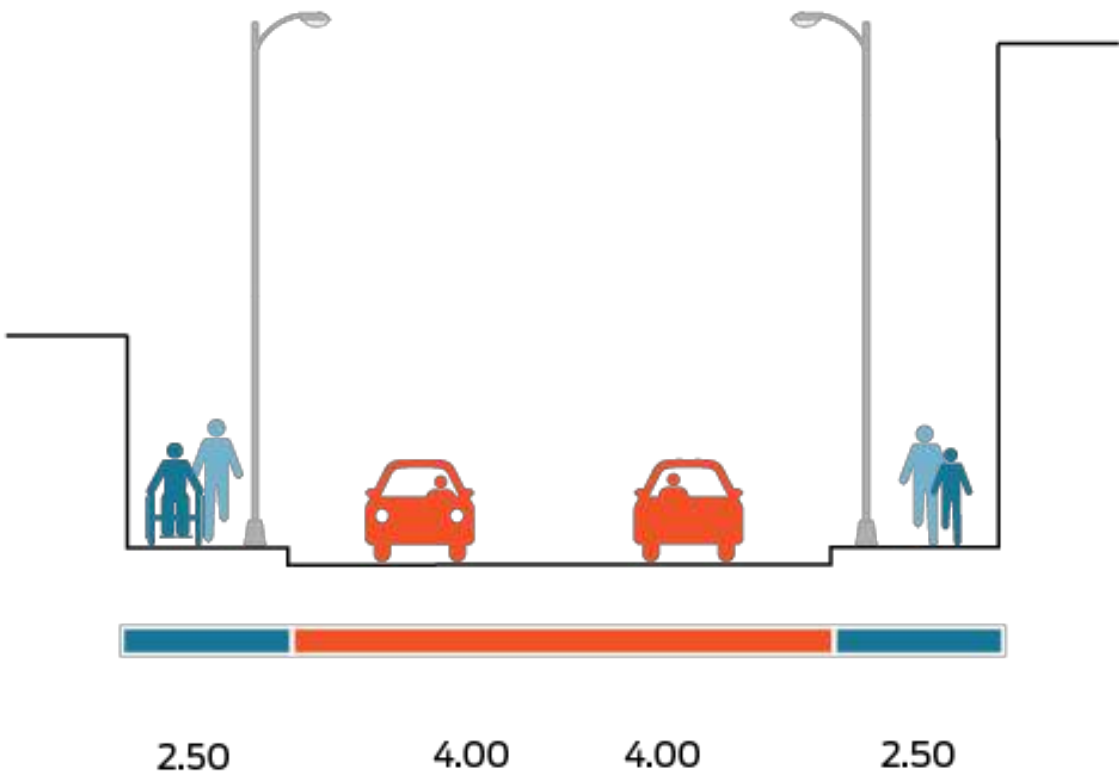


Connector Streets| Vision | After

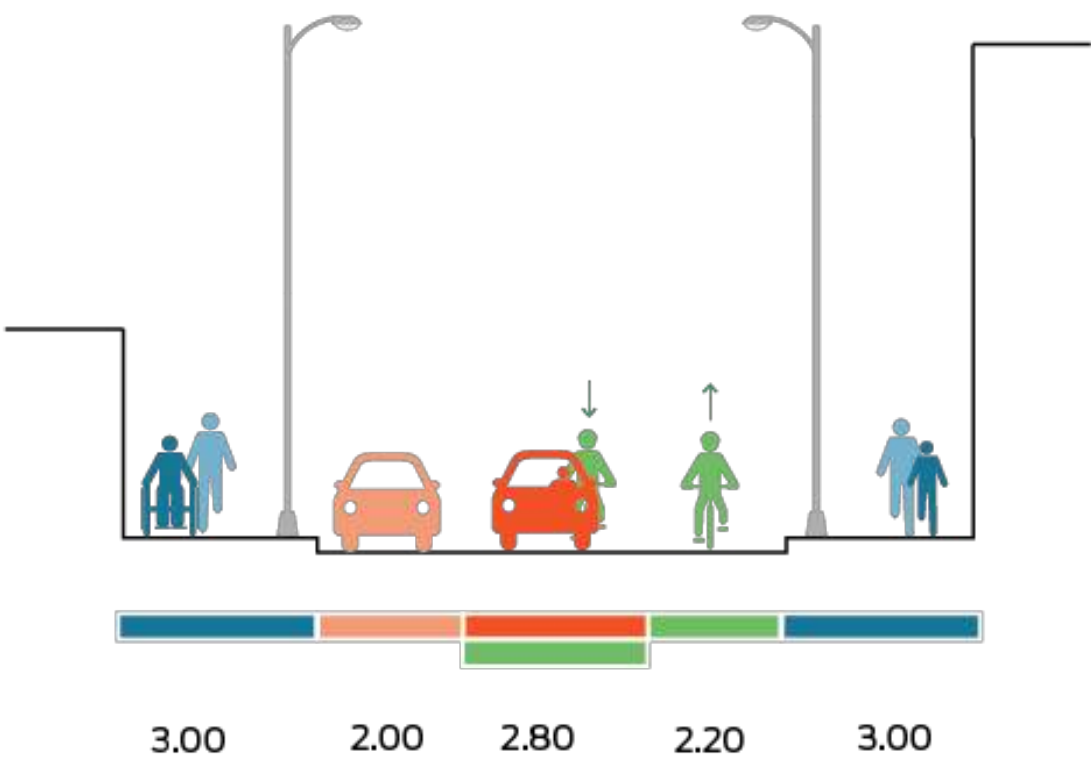
Addis Ababa | Ethiopia

Cycle corridors on Local Streets(<18m)

EXISTING:



PROPOSED:





Local Street| Vision | Before



Local Street| Vision | After

Addis Ababa | Ethiopia

Additional Considerations



Construction quality and maintenance



Curbside management and enforcement



Adequate pedestrian facilities to ensure safety for all



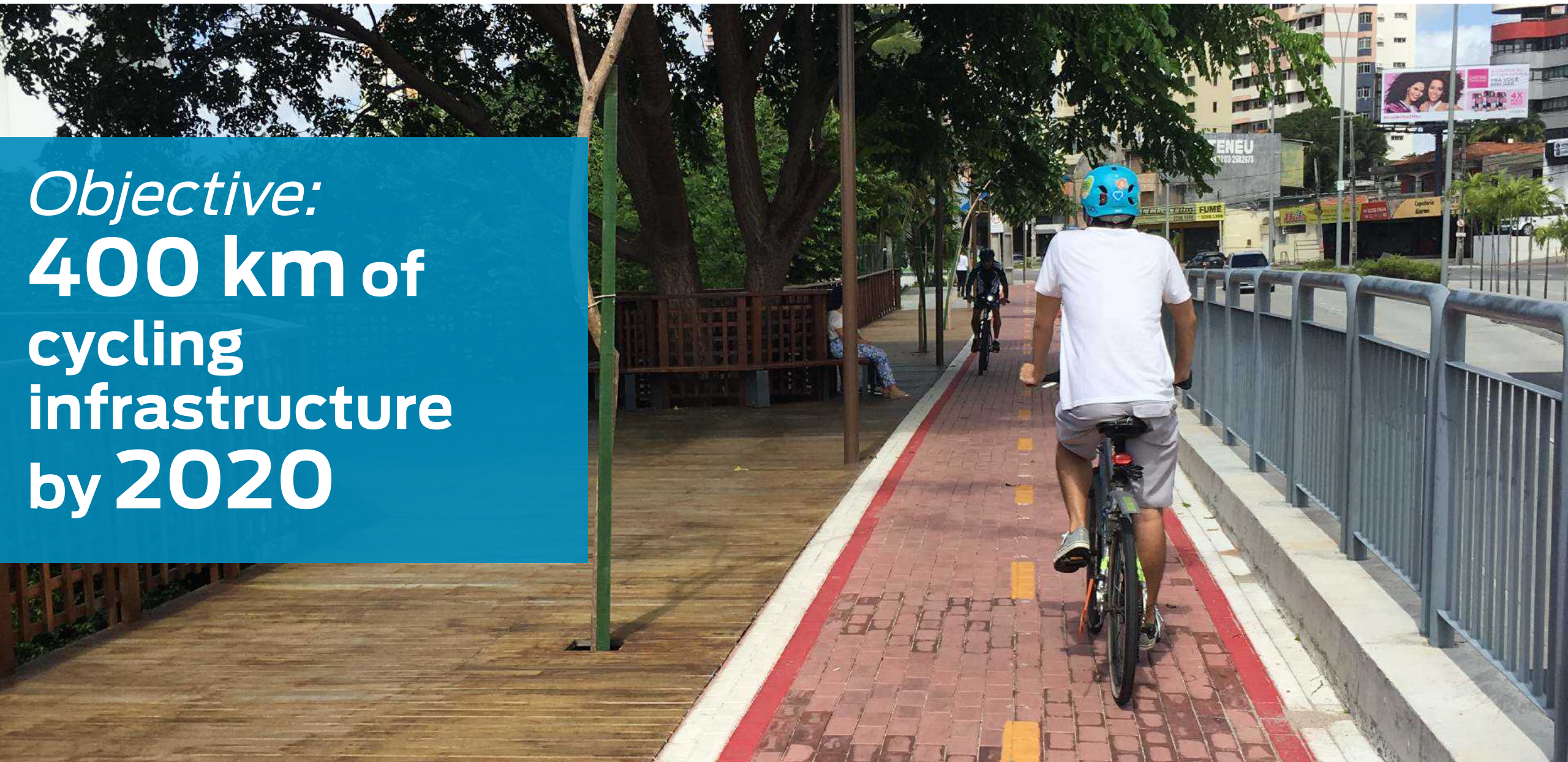
Vision:

By 2028

**5% mode share
by cycling**

Fortaleza

(Brazil)



Objective:
**400 km of
cycling
infrastructure
by 2020**



**INSTITUTIONAL
AND POLITICAL
WILL**



INFRASTRUCTURE



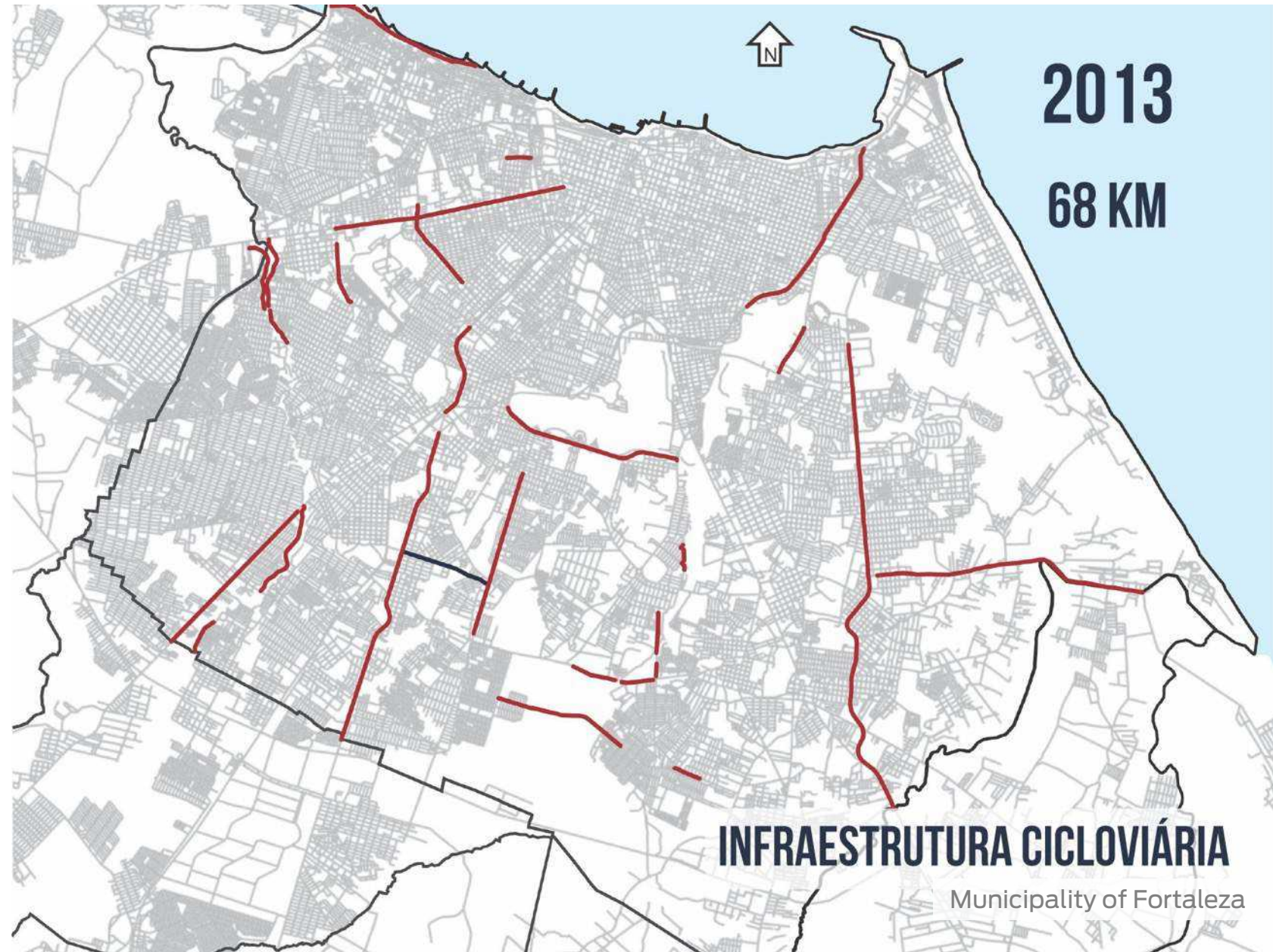
BIKE SHARING



**EDUCATION,
COMMUNICATION,
AND DATA**

Fortaleza | Brazil

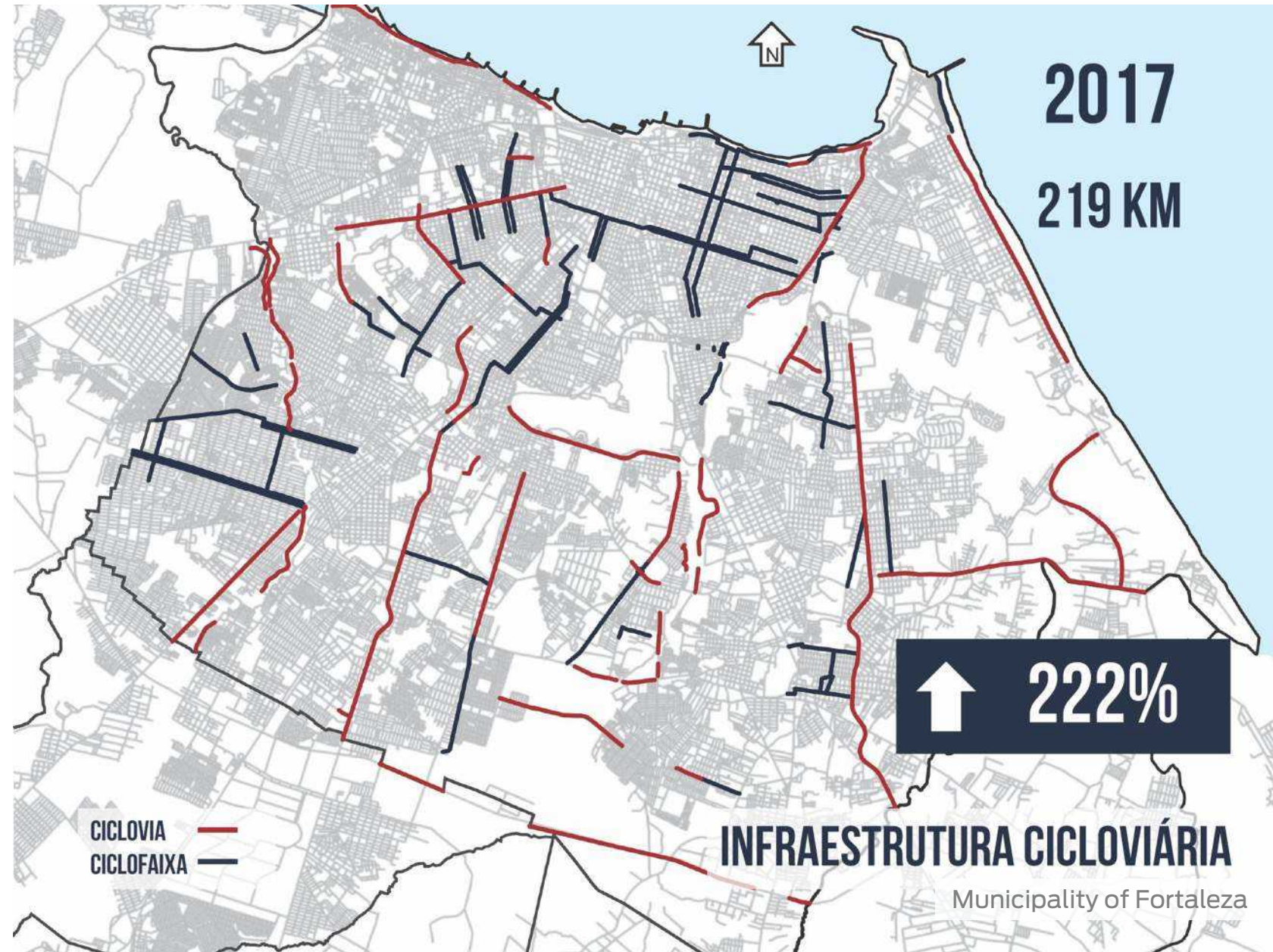
Bike infrastructure



Fortaleza | Brazil

Bike infrastructure

Increase of
222%
in 4 years



Fortaleza | Brazil

Bike infrastructure



Fortaleza | Brazil

Bike infrastructure



Fortaleza | Brazil

Bike infrastructure



NARROWING TRAFFIC LANES HAS THE
POTENCIAL TO REDUCE UP TO

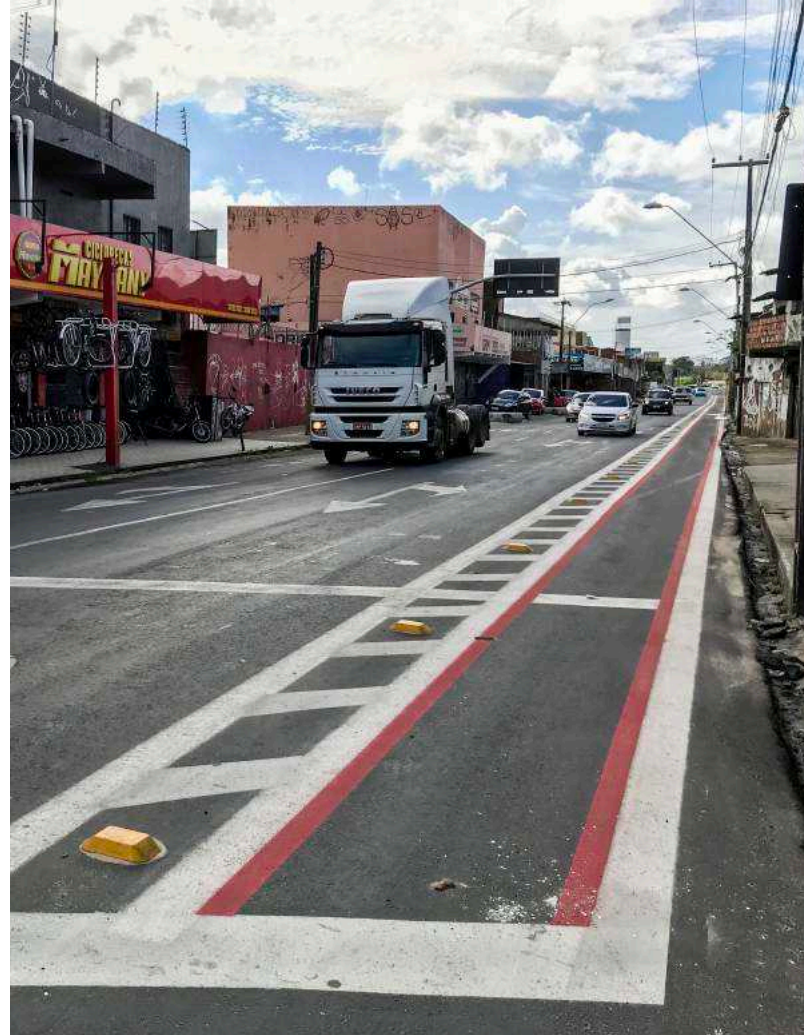
30% ROAD CRASHES
WITH VICTIMS

Safety Perception

Comparison - Three Scenarios



Unprotected Cycle Lane



Buffered Cycle Lane



Buffered Cycle Lane + Flex Posts

Intercept Surveys with Cyclists

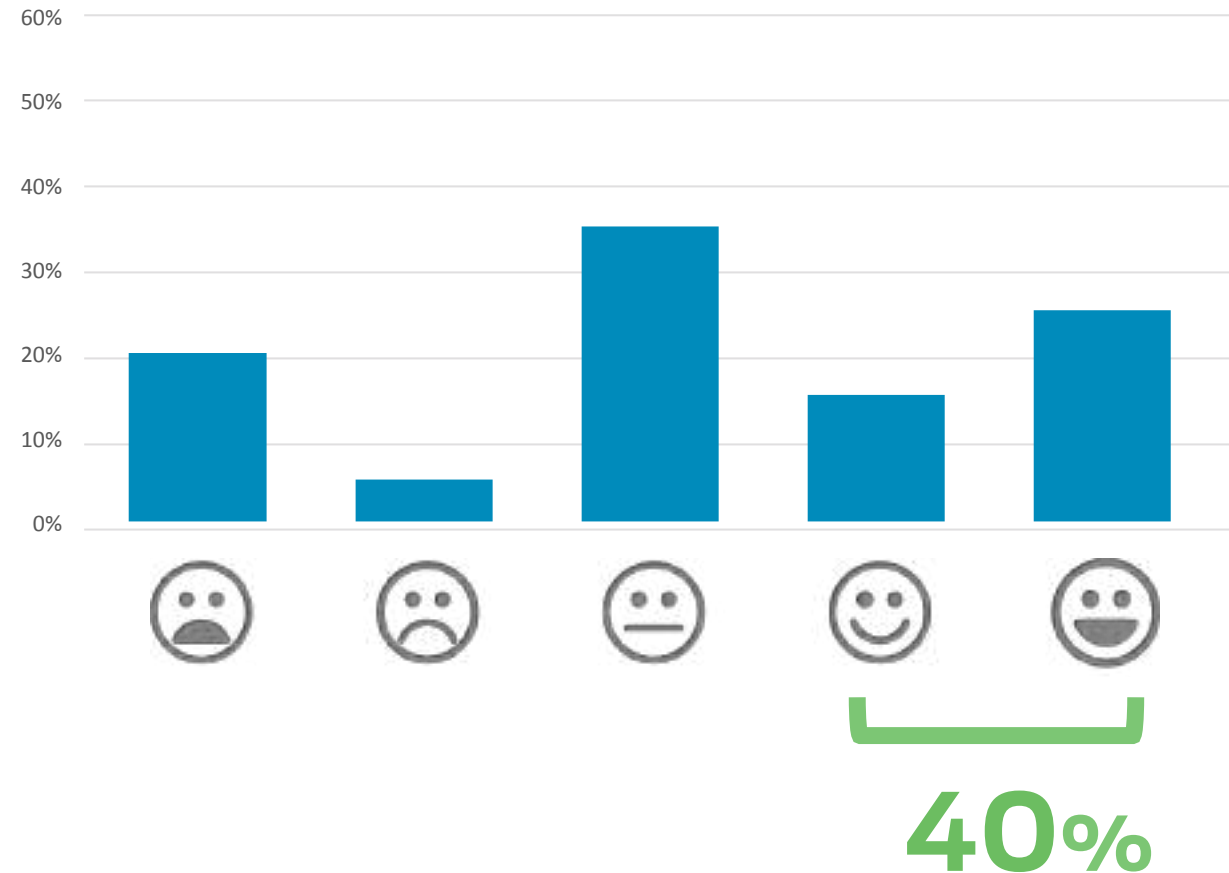


Conventional Cycle Lane

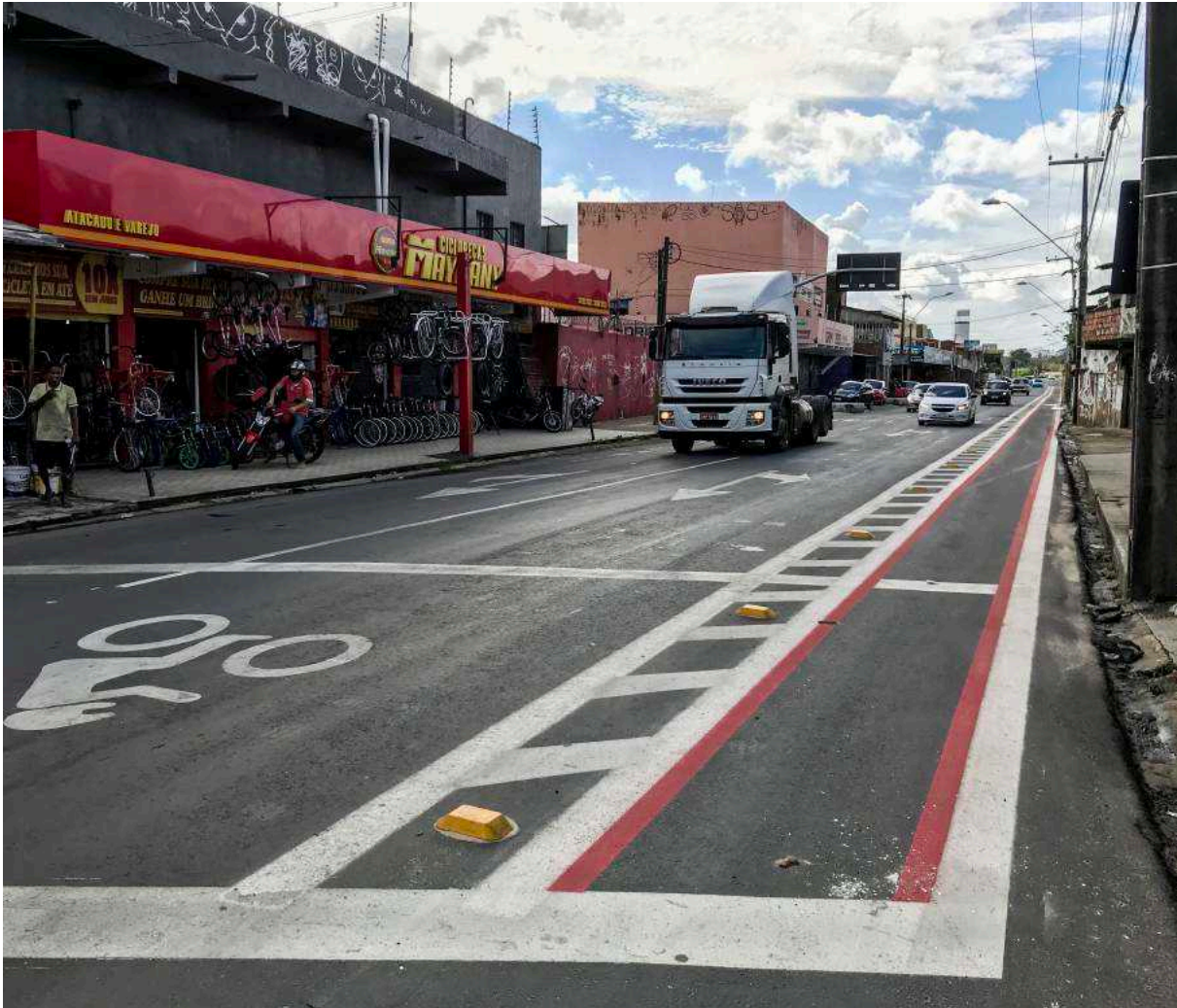


Raimundo Neri St.

How safe did you feel riding within the last block?

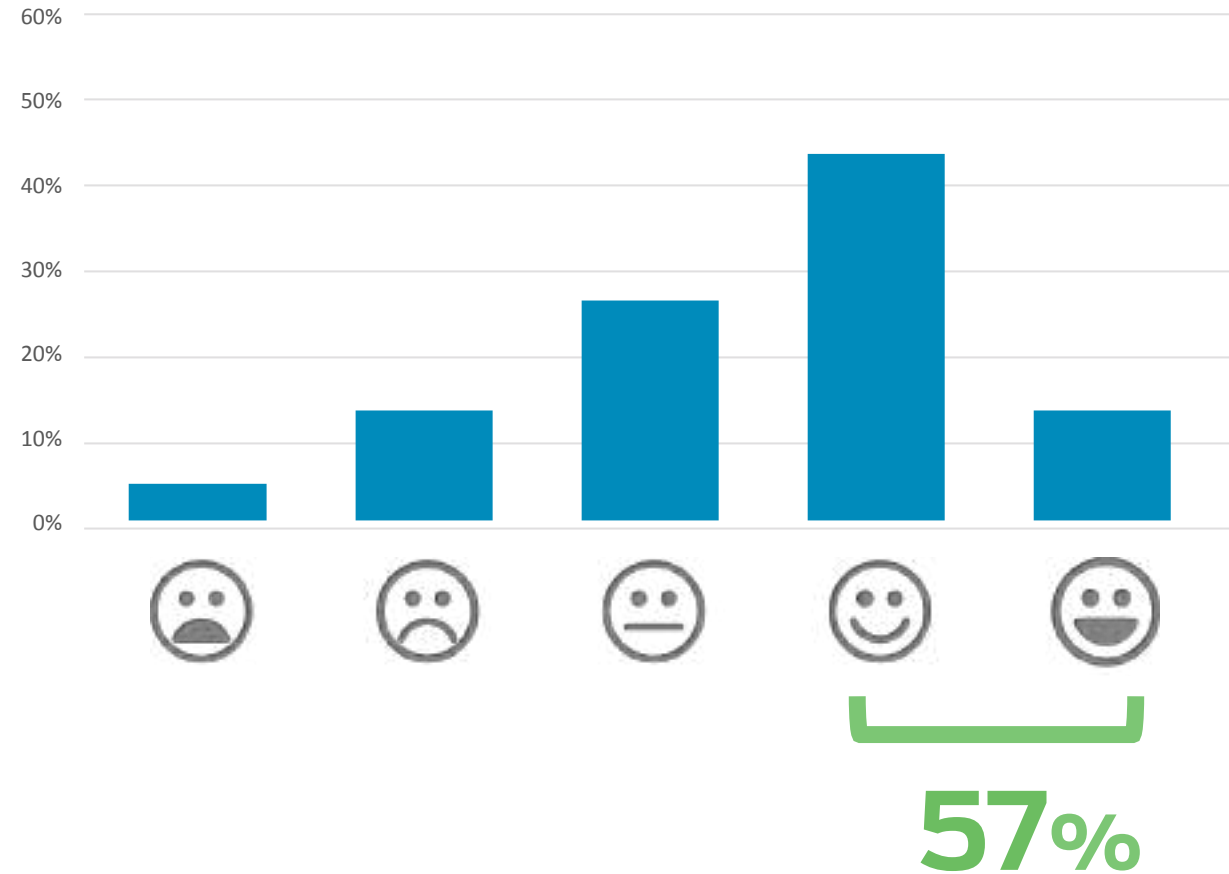


Buffered Cycle Lane



Luiz Vieira Ave.

How safe did you feel riding within the last block?



Buffered Cycle Lane + Flex Posts



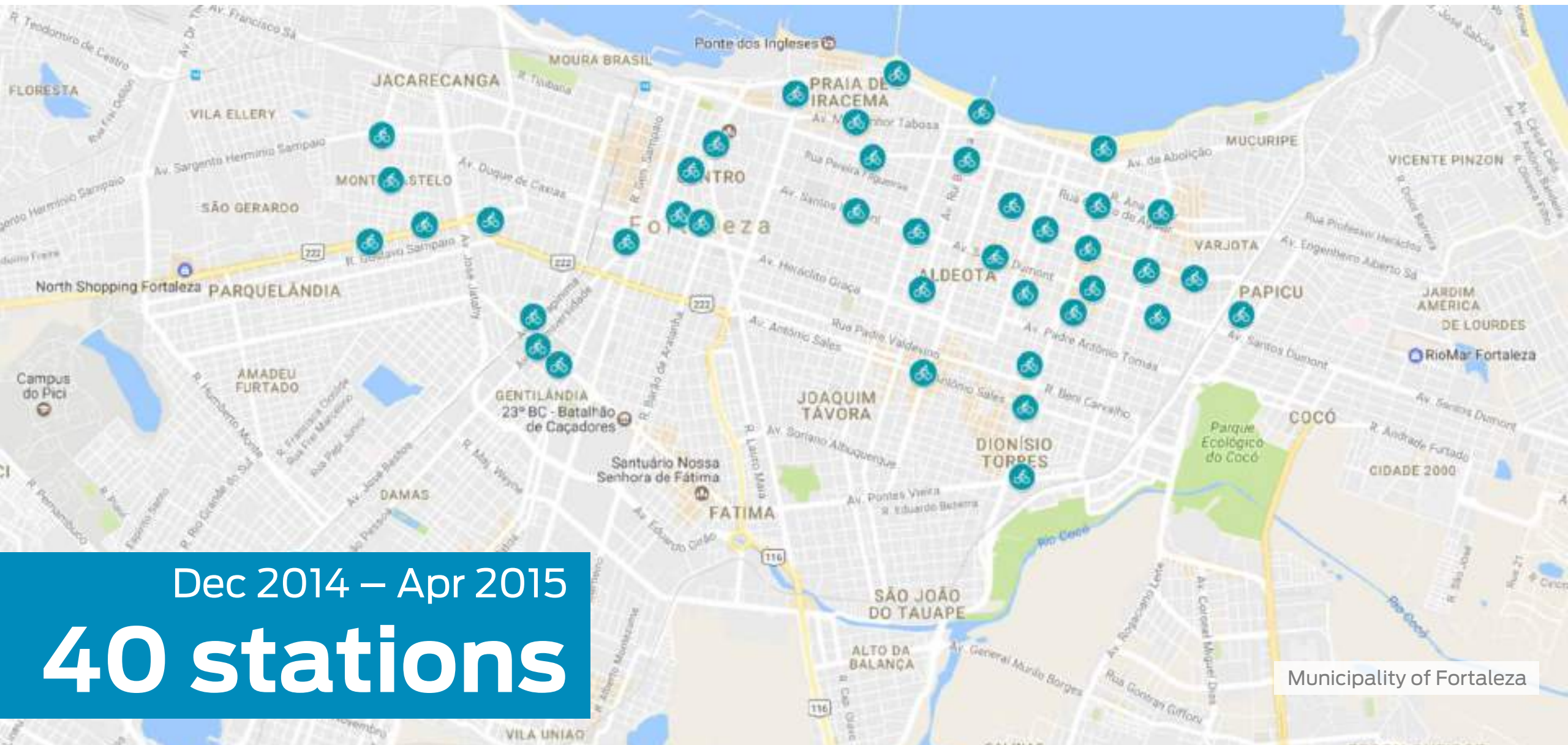
Av. Luiz Vieira

How safe did you feel riding within the last block?



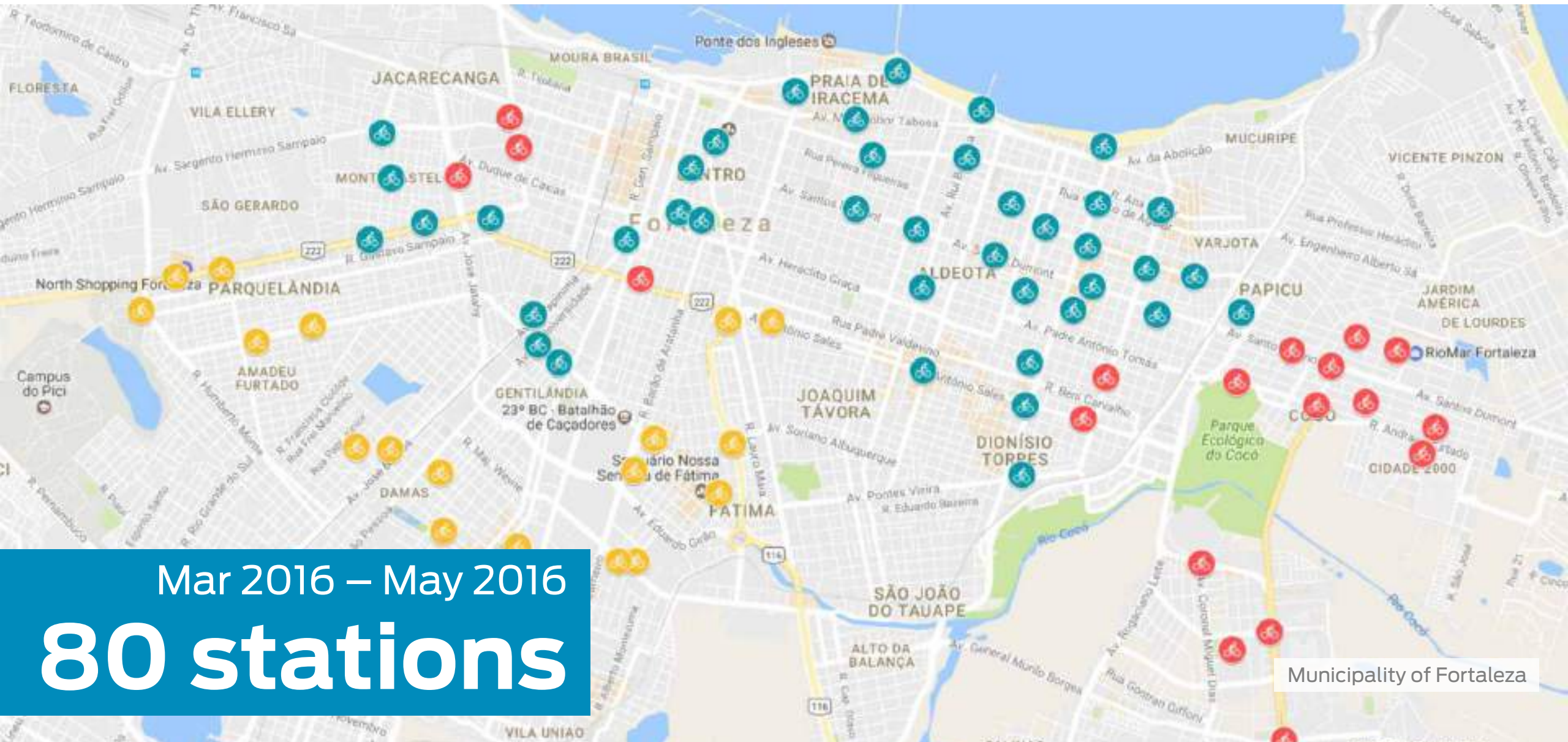
Fortaleza | Brazil

Bike sharing | Bicicletar



Fortaleza | Brazil

Bike sharing | Bicicletar



Mar 2016 – May 2016

80 stations

Municipality of Fortaleza



Integrated bike system



14 hours



Integrated
public transit
ticket

Register

Pick a bike

Drop the bike

Fortaleza | Brazil

Bike sharing + Education | Mini bicicletar



In 2017, more than
6,000 trips

Municipality of Fortaleza



From 2011 to 2016

39.5%

reduction

of crashes involving

cyclists



Vision:

The most
cyclable
city in Brazil

Bogotá

(Colombia)



Objective:
10% of total
trips by bicycle
by **2020**

Mobility Department of Bogotá

Bogotá | Colombia

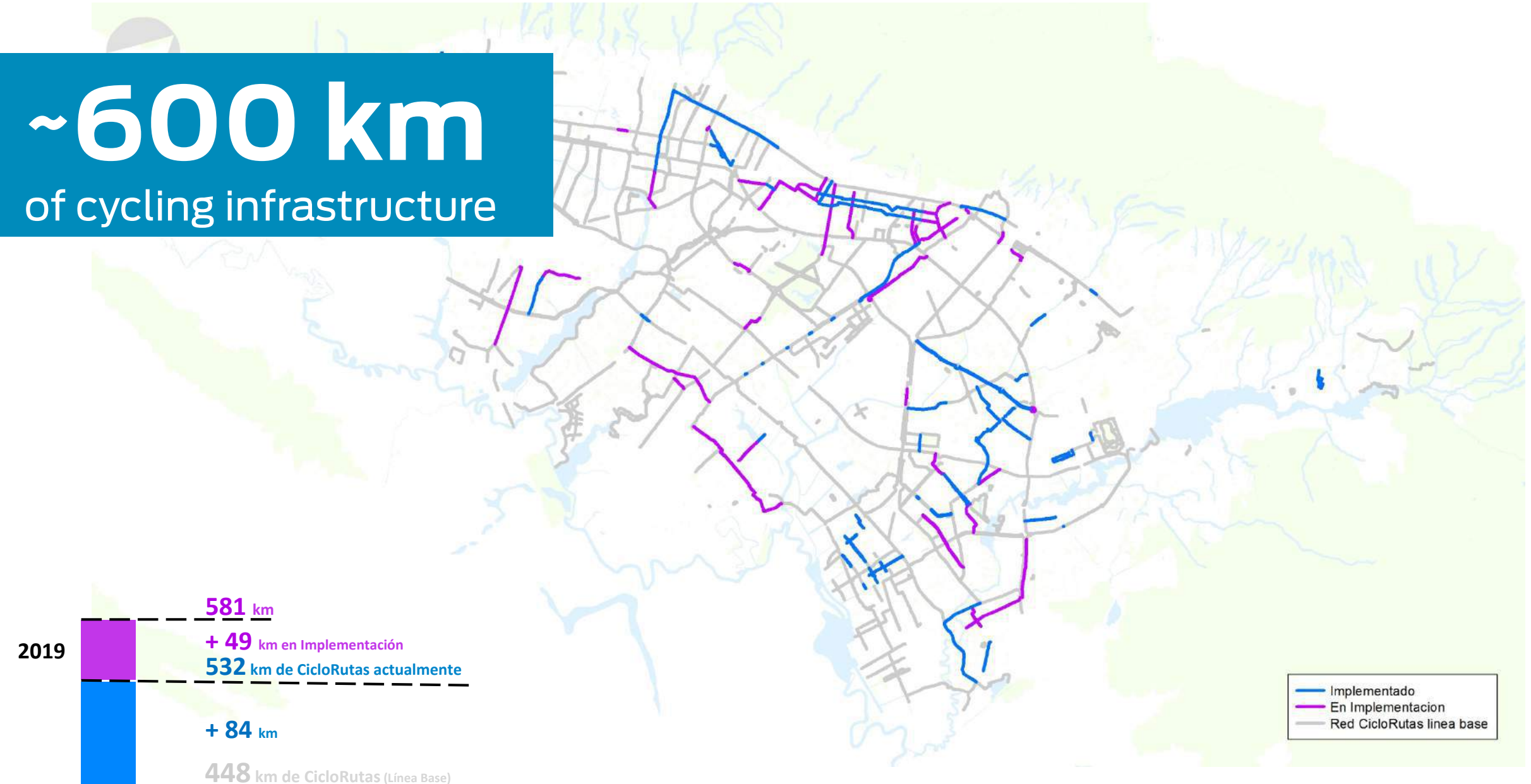
Evolution of the cycling network



Bogotá | Colombia

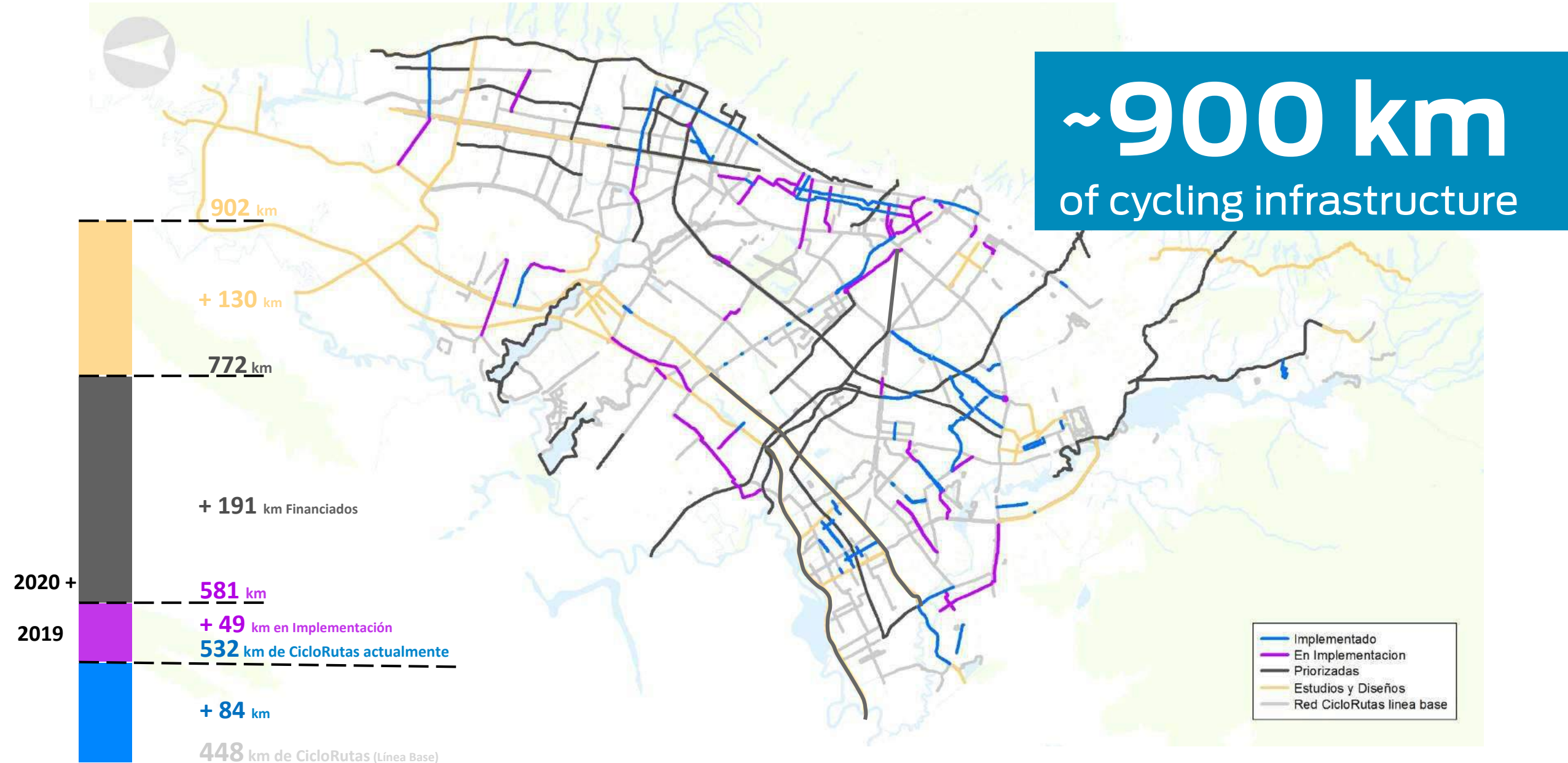
Evolution of the cycling network

~600 km
of cycling infrastructure



Bogotá | Colombia

Evolution of the cycling network



Bogotá | Colombia

Evolution of the cycling network





Ciclovía

up to 2 million people
125 km of open streets

Bike-to-school

4,000 children
90 schools
70+ routes

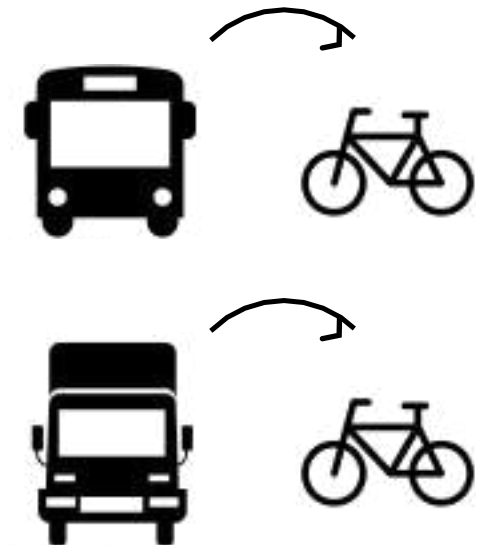




Cyclist manual



Role playing



Bike nursery

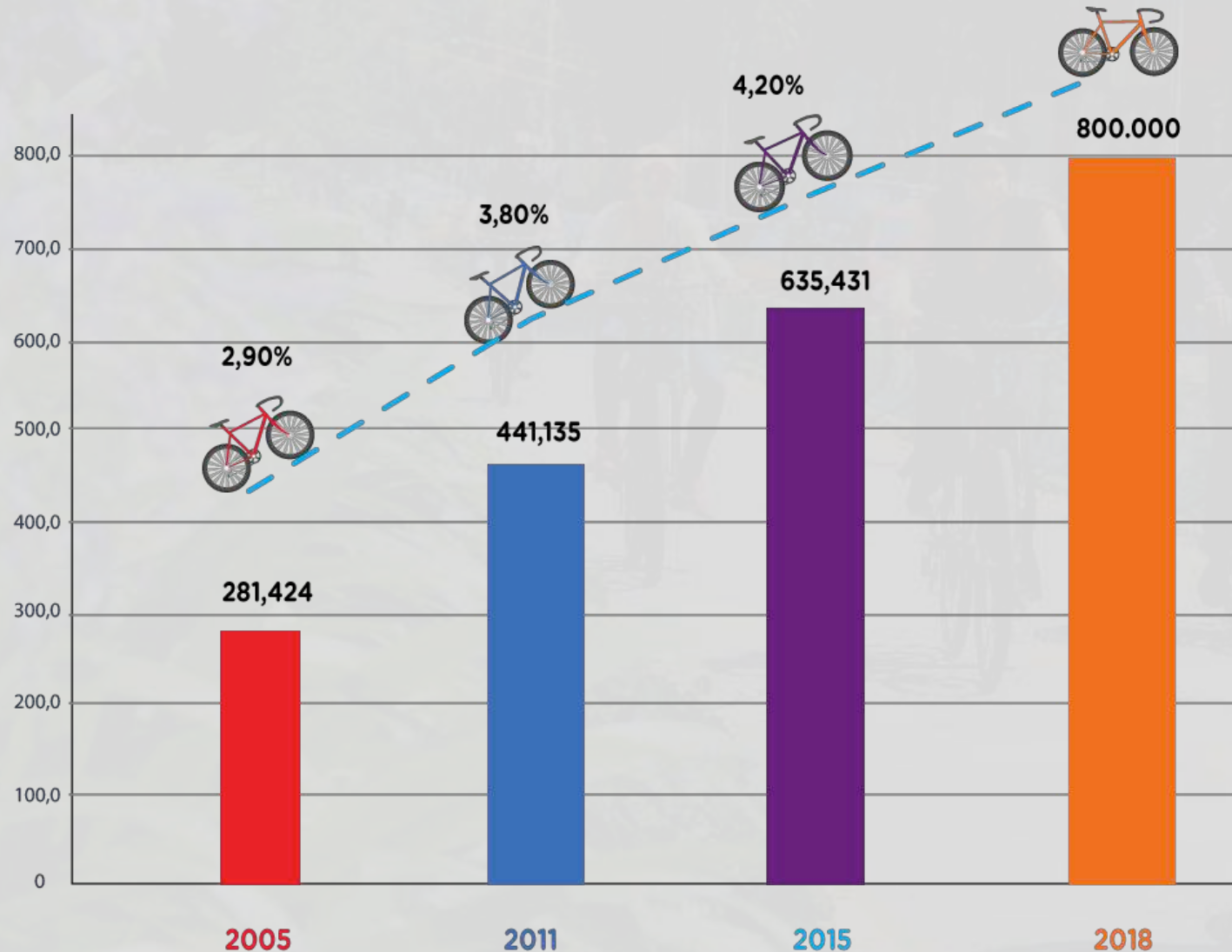




Bike week
26,000 attendees
59 events
69 organizations involved

Bogotá | Colombia

Increase of number of trips by bike



800,000
trips a day
6% of total ridership



In 2019

900,000

Trips per day by bike

Bogotá | Colombia

Increase of number of trips by bike

Long term
vision:

Bogotá
Capital Mundial
de la Bici



Addis Ababa Ethiopia



Fortaleza Brazil



Bogotá Colombia



**Emerging
Cycling
Movement**

**Progressing
Cycling
Momentum**

**Reinforcing
Cycling
Culture**

THANK YOU!

www.globaldesigningcities.org

**Fabrizio Prati and
Abhimanyu Prakash**



@GlobalStreets



facebook.com/globalstreets

